AWS
CERTIFIED

TECHSHIKSHA

Cloud Practitioner

CLF-C02

300
Interview Questions & Answers

Real-World Examples | All Exam Domains | 2025 Edition

EXAM DOMAINS COVERED

Cloud Concepts 24%
Exam Weight

Security & Compliance 30%
Exam Weight

Cloud Technology 34%
Exam Weight

Billing & Pricing 12%
Exam Weight

Exam Code	Questions	Duration	Pass Score
CLF-C02	65 (MCQ + MR)	90 Minutes	700 / 1000

▯ Table of Contents

Total: 300 Questions across 14 Topic Sections

Section 1 · Cloud Concepts

Q1 What is Cloud Computing?

A Cloud computing is the on-demand delivery of IT resources — servers, storage, databases, networking, software — over the internet ('the cloud') with pay-as-you-go pricing. Instead of buying and owning physical data centers, you access these services from AWS.

▯ *Real-World Example: A startup uses AWS S3 to store user-uploaded photos instead of buying physical servers. They pay only for the gigabytes stored each month.*

Q2 What are the three main cloud service models?

A 1. IaaS (Infrastructure as a Service) – provides virtualised computing resources (EC2, S3). 2. PaaS (Platform as a Service) – provides a platform to build apps without managing infrastructure (Elastic Beanstalk). 3. SaaS (Software as a Service) – fully managed software delivered over the web (Amazon WorkMail).

▯ *Real-World Example: A developer uses Elastic Beanstalk (PaaS) to deploy a Python Flask app. He uploads code and AWS manages servers, load balancers, and scaling automatically.*

Q3 What are the three cloud deployment models?

A 1. Public Cloud – resources owned and operated by AWS, shared among customers. 2. Private Cloud – dedicated infrastructure for one organisation (on-premises or hosted). 3. Hybrid Cloud – combination of public and private cloud connected via VPN or AWS Direct Connect.

▯ *Real-World Example: A bank keeps sensitive customer data on a private cloud but uses AWS for web-facing apps — this is a hybrid cloud architecture.*

Q4 What are the 6 advantages of cloud computing according to AWS?

A 1. Trade capital expense for variable expense. 2. Benefit from massive economies of scale. 3. Stop guessing capacity. 4. Increase speed and agility. 5. Stop spending money on running and maintaining data centers. 6. Go global in minutes.

▯ *Real-World Example: A company avoids buying $1M in servers (CapEx) and instead pays $5,000/month on AWS (OpEx), converting a fixed cost to a variable cost.*

Q5 What is the AWS Shared Responsibility Model?

A AWS is responsible for security 'of' the cloud (physical hardware, global infrastructure, managed services). Customers are responsible for security 'in' the cloud (data encryption, IAM policies, OS patching on EC2, firewall rules).

▯ *Real-World Example: AWS secures its data center buildings and hypervisors. The customer must configure security groups, encrypt S3 buckets, and manage IAM permissions for their employees.*

Q6 What is a Region in AWS?

A An AWS Region is a physical geographic area containing multiple, isolated Availability Zones (data centers). Each region is completely independent. You choose a region based on latency, data residency requirements, and service availability. Examples: us-east-1 (N. Virginia), ap-south-1 (Mumbai).

▯ *Real-World Example: An Indian company chooses the ap-south-1 Mumbai region to keep data within India for compliance with data-localisation laws and to achieve low latency for Indian users.*

Q7 What is an Availability Zone (AZ)?

A An Availability Zone is one or more discrete data centers within an AWS Region, each with redundant power, networking, and connectivity. AZs within a region are interconnected with high-bandwidth, low-latency networking. Deploying across multiple AZs provides high availability.

▯ *Real-World Example: An application runs on EC2 instances in us-east-1a and us-east-1b. If one AZ loses power, the other AZ continues serving traffic without downtime.*

Q8 What is an AWS Edge Location?

A Edge Locations are AWS data center endpoints used by Amazon CloudFront (CDN) and Route 53. They cache content closer to end users to reduce latency. There are many more edge locations than regions (400+ globally).

▯ *Real-World Example: A US user requests a video from a company hosted in Mumbai. CloudFront serves it from a New York edge location, reducing load time from 250ms to 15ms.*

Q9 What is the AWS Global Infrastructure?

A AWS global infrastructure consists of Regions, Availability Zones, and Edge Locations. As of 2024, AWS has 33 launched regions, 105 AZs, and 400+ edge locations globally, providing low latency, fault tolerance, and compliance coverage worldwide.

Q10	What is High Availability vs Fault Tolerance?
A	High Availability (HA) means a system is designed to remain operational with minimal downtime — typically achieved by deploying across multiple AZs with auto-scaling. Fault Tolerance means a system continues operating even when components fail, often requiring redundancy at every layer.
▯	*Real-World Example: HA: An ELB distributes traffic to 2 EC2 instances across 2 AZs — if one fails, the other handles traffic. Fault Tolerance: A storage system that writes data to 3 nodes simultaneously so even if 1 fails, no data is lost.*

Q11	What is Elasticity in cloud computing?
A	Elasticity is the ability to automatically scale resources up or down based on demand. AWS Auto Scaling achieves this — it adds EC2 instances when traffic spikes and removes them when demand drops, ensuring cost efficiency.
▯	*Real-World Example: An e-commerce site uses Auto Scaling to run 2 EC2 instances normally. During a Diwali sale, it automatically scales to 20 instances to handle 10x traffic, then scales back down afterwards.*

Q12	What is Scalability?
A	Scalability is the ability to handle increased load by adding resources. Vertical scaling (scale-up) adds more power to an existing instance (larger EC2 type). Horizontal scaling (scale-out) adds more instances. Cloud supports both, but horizontal scaling is preferred for resilience.
▯	*Real-World Example: A database server upgraded from db.t3.medium to db.r5.4xlarge = vertical scaling. Adding 5 more EC2 app servers behind a load balancer = horizontal scaling.*

Q13	What is the difference between CapEx and OpEx?
A	CapEx (Capital Expenditure) is upfront investment in physical infrastructure (servers, buildings) that depreciates over time. OpEx (Operational Expenditure) is ongoing operational costs paid as you use them. Cloud computing converts CapEx to OpEx, improving cash flow and flexibility.
▯	*Real-World Example: Traditional IT: Buy a $500K server (CapEx). Cloud: Pay $2,000/month for equivalent EC2 capacity (OpEx). The company can reinvest the $500K in the business.*

Q14	What is a Cloud-Native application?

A	A cloud-native application is designed specifically to run in the cloud, taking full advantage of cloud services like auto-scaling, managed databases, serverless functions, and microservices. It is typically containerised, follows the 12-factor app methodology, and uses CI/CD pipelines.
▯	*Real-World Example: Netflix is cloud-native: it uses AWS microservices, auto-scales to millions of concurrent users, deploys hundreds of times per day, and has no single point of failure.*

Q15	What does 'Pay-as-you-go' pricing mean?
A	Pay-as-you-go means you only pay for the exact resources you consume — no upfront costs, no long-term contracts (in the on-demand model). You are billed by the second (EC2), GB stored (S3), or requests made (Lambda). This eliminates waste from over-provisioning.
▯	*Real-World Example: An EC2 t3.micro instance costs ~$0.0104/hour. If you run it for exactly 73 hours this month, you pay ~$0.76. Traditional hosting charges a flat monthly fee regardless of usage.*

Q16	What is the AWS Well-Architected Framework?
A	The AWS Well-Architected Framework provides best practices for building secure, high-performing, resilient, and efficient infrastructure. It has 6 pillars: 1. Operational Excellence, 2. Security, 3. Reliability, 4. Performance Efficiency, 5. Cost Optimization, 6. Sustainability.
▯	*Real-World Example: A company uses the AWS Well-Architected Tool to review their architecture. It identifies that they have no automated backups (Reliability gap) and are running oversized instances (Cost Optimization gap).*

Q17	What is the Reliability pillar of the Well-Architected Framework?
A	The Reliability pillar focuses on a system's ability to recover from failures, dynamically acquire resources to meet demand, and mitigate disruptions. Key practices: multi-AZ deployments, automated backups, Route 53 health checks, and chaos engineering.

Q18	What is the Security pillar of the Well-Architected Framework?
A	The Security pillar covers protecting information and systems. Key best practices: implement strong identity (IAM MFA), enable traceability (CloudTrail), apply security at all layers (NACLs + Security Groups), automate security responses, and protect data in transit and at rest.

Q19 What is the Cost Optimization pillar?

A The Cost Optimization pillar focuses on avoiding unnecessary costs. Best practices: right-size instances, use Reserved Instances or Savings Plans for steady-state workloads, delete unused resources, use S3 lifecycle policies, and analyse costs with Cost Explorer.

 Real-World Example: A company finds 40 unused EBS volumes and 15 idle EC2 instances via AWS Cost Explorer. Deleting them saves $3,000/month — a quick Cost Optimization win.

Q20 What is the Operational Excellence pillar?

A Operational Excellence is about running and monitoring systems to deliver business value and continually improving processes. Key practices: infrastructure as code (CloudFormation), frequent small reversible changes, anticipate failure, learn from operational failures.

Section 2 · AWS Core Services – Compute

Q21 What is Amazon EC2?

A Amazon Elastic Compute Cloud (EC2) provides resizable virtual servers (instances) in the cloud. You choose the OS, instance type (CPU/RAM), storage, and networking. EC2 instances are charged by the second. Use cases: web servers, application servers, batch processing.

 Real-World Example: A company runs their Java Spring Boot application on an EC2 t3.medium instance (2 vCPU, 4GB RAM) in ap-south-1. When traffic grows, they resize to t3.xlarge without buying new hardware.

Q22 What are EC2 instance families?

A EC2 families are optimised for different use cases: General Purpose (t3, m5) – balanced CPU/memory; Compute Optimised (c5) – high-performance processors; Memory Optimised (r5, x1) – large in-memory datasets; Storage Optimised (i3, d2) – high I/O; Accelerated Computing (p3, g4) – GPU workloads.

 Real-World Example: A machine learning team uses p3.8xlarge (4 NVIDIA Tesla V100 GPUs) for model training and m5.xlarge for their web application — different instance families for different needs.

Q23 What are EC2 pricing models?

A	1. On-Demand – pay by the second, no commitment, highest per-hour rate. 2. Reserved Instances – 1 or 3-year commitment, up to 72% discount. 3. Spot Instances – bid for unused capacity, up to 90% discount but can be interrupted. 4. Dedicated Hosts – physical server dedicated to you, for licensing compliance.
▯	*Real-World Example: A company's web servers run 24/7 — they buy 1-year Reserved Instances (saving 40%). For batch data processing jobs, they use Spot Instances (saving 80%) since interruptions are acceptable.*

Q24	What is EC2 Auto Scaling?
A	EC2 Auto Scaling automatically adds or removes EC2 instances based on defined policies. You set minimum, maximum, and desired capacity. Scaling policies can be based on CPU utilisation, custom CloudWatch metrics, or scheduled events. It maintains availability and optimises costs.
▯	*Real-World Example: During business hours (9AM–6PM), a company schedules Auto Scaling to maintain 10 EC2 instances. At night, it scales down to 2, saving 80% on compute costs.*

Q25	What is an Amazon Machine Image (AMI)?
A	An AMI is a template that contains the software configuration (OS, application server, applications) required to launch an EC2 instance. AWS provides public AMIs (Amazon Linux, Ubuntu, Windows Server). You can create custom AMIs to standardise deployments.
▯	*Real-World Example: A DevOps team creates a custom AMI with Apache, Java 11, and their application pre-installed. They use this AMI to launch 50 identical EC2 instances across an Auto Scaling group within minutes.*

Q26	What is AWS Lambda?
A	AWS Lambda is a serverless compute service that runs your code in response to events without provisioning or managing servers. You pay only for the compute time consumed (billed per millisecond). Supports Node.js, Python, Java, Go, etc. Max execution time: 15 minutes.
▯	*Real-World Example: An image-upload API triggers a Lambda function whenever a photo is saved to S3. Lambda resizes the image to 3 sizes and saves thumbnails back to S3. No servers to manage, cost scales to zero when no uploads occur.*

Q27	What is AWS Elastic Beanstalk?

A	Elastic Beanstalk is a PaaS service that automatically handles deployment, capacity provisioning, load balancing, auto-scaling, and health monitoring. You simply upload your code (Java, .NET, Python, Node.js, etc.) and AWS manages the rest.
▯	*Real-World Example: A developer uploads a Django web app ZIP to Elastic Beanstalk. AWS automatically creates an EC2 instance, sets up the load balancer, configures security groups, and deploys the app — all in 5 minutes.*

Q28	What is Amazon ECS?
A	Amazon Elastic Container Service (ECS) is a fully managed container orchestration service. It runs Docker containers on a cluster of EC2 instances or AWS Fargate (serverless). ECS handles scheduling, scaling, and monitoring of containers.
▯	*Real-World Example: A microservices application has 8 Docker containers (auth, products, orders, etc.). ECS deploys each as a task, auto-scales each service independently, and restarts failed containers automatically.*

Q29	What is AWS Fargate?
A	AWS Fargate is a serverless compute engine for containers (ECS and EKS). You define the container and its CPU/memory requirements. Fargate provisions and manages the underlying infrastructure. You don't manage EC2 instances.
▯	*Real-World Example: A team deploys a containerised Node.js API on ECS with Fargate. They specify 0.5 vCPU and 1GB RAM per container. AWS handles server provisioning — the team only writes application code.*

Q30	What is Amazon EKS?
A	Amazon Elastic Kubernetes Service (EKS) is a managed Kubernetes service. AWS manages the Kubernetes control plane (API server, etcd). You manage worker nodes (EC2 or Fargate). EKS is ideal for teams already using Kubernetes.
▯	*Real-World Example: A large company already runs Kubernetes on-premises. They lift-and-shift their Kubernetes workloads to EKS, maintaining the same kubectl commands and Helm charts while AWS manages the control plane.*

Q31	What is AWS Batch?

A	AWS Batch enables developers to run batch computing jobs on AWS. It dynamically provisions EC2 or Spot instances, schedules and runs jobs, and scales compute automatically. Ideal for high-performance computing, genomics, financial modelling.
▯	*Real-World Example: A pharmaceutical company runs 10,000 molecular simulation jobs every night. AWS Batch automatically launches Spot EC2 instances, distributes jobs, and terminates instances when complete — saving 80% vs on-demand.*

Q32	What is Amazon Lightsail?
A	Amazon Lightsail is a simplified cloud platform designed for developers, small businesses, and students who need a straightforward, low-cost way to build websites and apps. It bundles compute, SSD storage, data transfer, DNS, and a static IP at a fixed monthly price.
▯	*Real-World Example: A blogger sets up a WordPress site on Lightsail for $5/month. They get a pre-configured Linux instance, 20GB SSD, 1TB data transfer, and a static IP — no complex AWS configuration required.*

Q33	What is the difference between EC2 and Lambda?
A	EC2 is a virtual machine you provision and manage, billed per second, always running. Lambda is serverless — runs only when triggered, billed per millisecond of execution, scales automatically to thousands of concurrent executions. Lambda is ideal for short, event-driven tasks; EC2 for long-running, stateful applications.
▯	*Real-World Example: EC2: Run a web server 24/7 for consistent traffic. Lambda: Process a payment webhook that fires 1,000 times/day — no server running when there are no events.*

Q34	What is EC2 User Data?
A	EC2 User Data is a script that runs automatically when an EC2 instance first starts. It's used to install software, update OS packages, configure services, or perform any bootstrap actions. Scripts must start with #!/bin/bash (for Linux).
▯	*Real-World Example: User Data script: #!/bin/bash / yum update -y / yum install -y httpd / systemctl start httpd — this automatically installs and starts Apache web server on every new EC2 instance.*

Q35	What is an EC2 Security Group?

A	A Security Group is a virtual firewall for EC2 instances that controls inbound and outbound traffic. Rules specify protocol, port range, and source/destination. Security groups are stateful — if you allow inbound traffic, the return traffic is automatically allowed. Default: all outbound allowed, all inbound denied.
▯	*Real-World Example: A web server security group: Allow inbound port 80 (HTTP) from 0.0.0.0/0, port 443 (HTTPS) from 0.0.0.0/0, port 22 (SSH) from your company IP only (203.0.113.0/32).*

Section 3 · Storage Services

Q36	What is Amazon S3?
A	Amazon Simple Storage Service (S3) is object storage offering unlimited capacity, 99.999999999% (11 nines) durability, and 99.99% availability. Objects are stored in buckets. S3 supports versioning, lifecycle policies, static website hosting, and event notifications. S3 is region-specific but bucket names must be globally unique.
▯	*Real-World Example: Airbnb stores millions of property photos in S3. Each photo is an object. S3 automatically replicates data across 3+ AZs within a region, ensuring durability even if two AZs fail simultaneously.*

Q37	What are S3 Storage Classes?
A	S3 Standard – frequent access, highest cost. S3 Intelligent-Tiering – auto-moves objects between tiers. S3 Standard-IA (Infrequent Access) – lower cost, retrieval fee. S3 One Zone-IA – single AZ, cheaper. S3 Glacier Instant Retrieval – archives, milliseconds retrieval. S3 Glacier Flexible Retrieval – hours retrieval. S3 Glacier Deep Archive – lowest cost, 12-hour retrieval.
▯	*Real-World Example: A company stores compliance documents accessed daily in S3 Standard, monthly audit reports in S3 Standard-IA (30% cheaper), and 7-year legal archives in S3 Glacier Deep Archive (90% cheaper than Standard).*

Q38	What is an S3 Bucket Policy?
A	An S3 Bucket Policy is a resource-based IAM policy attached to an S3 bucket. It defines who (AWS accounts, IAM users, roles) can access the bucket and what actions they can perform. Written in JSON. Used for cross-account access and making buckets public.

▯	*Real-World Example: A public website hosted on S3 needs a bucket policy: {Effect: Allow, Principal: *, Action: s3:GetObject, Resource: arn:aws:s3:::my-website/*} — this makes all objects publicly readable.*

Q39	What is S3 Versioning?
A	S3 Versioning keeps multiple versions of an object. When enabled, every time an object is overwritten or deleted, the previous version is preserved. You can restore previous versions. Versioning protects against accidental deletes and overwrites. Once enabled, versioning cannot be disabled (only suspended).
▯	*Real-World Example: A developer accidentally overwrites a critical config.json file. With versioning enabled, they can retrieve the previous version from the S3 console in seconds, preventing data loss.*

Q40	What is Amazon EBS?
A	Amazon Elastic Block Store (EBS) provides persistent block-level storage volumes for EC2 instances. Like a virtual hard drive. EBS volumes persist independently from EC2 instance lifecycle. Types: gp3 (general purpose SSD), io2 (high-performance SSD), st1 (throughput optimised HDD), sc1 (cold HDD).
▯	*Real-World Example: An EC2 instance running MySQL uses a 500GB gp3 EBS volume for the database files. If the EC2 instance is terminated, the EBS volume persists and can be attached to a new EC2 instance — data is not lost.*

Q41	What is Amazon EFS?
A	Amazon Elastic File System (EFS) is a managed NFS file system that can be mounted by multiple EC2 instances simultaneously (shared file system). It automatically scales from gigabytes to petabytes. EFS is Regional — it spans multiple AZs. Ideal for shared workloads, content management, big data.
▯	*Real-World Example: A media company has 20 EC2 instances that all need to read/write the same video files. EFS allows all 20 instances to mount the same file system concurrently — impossible with EBS (which is attached to one EC2).*

Q42	What is Amazon S3 Glacier?
A	S3 Glacier is a low-cost object storage designed for data archiving. Three retrieval options: Instant (milliseconds), Flexible (minutes to hours), Deep Archive (12 hours). Deep Archive costs ~$1/TB/month. Ideal for compliance archives, backup data, raw media assets.

▯	*Real-World Example: A healthcare company must retain patient records for 10 years by law. They store them in S3 Glacier Deep Archive at $1/TB/month — 95% cheaper than S3 Standard for data they hope never to access.*

Q43	What is AWS Storage Gateway?
A	AWS Storage Gateway is a hybrid cloud storage service that connects on-premises environments to AWS cloud storage. Types: S3 File Gateway (stores files as S3 objects), Volume Gateway (iSCSI block storage), Tape Gateway (virtual tape library). Used for backup and disaster recovery.
▯	*Real-World Example: A company's backup software already writes to tape drives. They replace physical tapes with a Tape Gateway — backup data goes to S3 Glacier automatically, eliminating tape management and offsite transport.*

Q44	What is the difference between EBS and S3?
A	EBS is block storage — low-latency, mounted to one EC2 at a time, used for OS, databases, apps. S3 is object storage — internet-accessible, unlimited scale, used for files, media, backups, static websites. EBS requires an EC2 instance; S3 is accessed via HTTPS API.
▯	*Real-World Example: Database files (MySQL data directory) ▯ use EBS. Profile pictures uploaded by users ▯ use S3. Both serve different purposes in the same application architecture.*

Q45	What is AWS Snowball?
A	AWS Snowball is a physical data migration device. AWS ships you a rugged appliance; you load your data onto it (up to 80TB per device), ship it back to AWS, and AWS uploads it to S3. Used when network bandwidth is insufficient for large data migrations.
▯	*Real-World Example: A company needs to migrate 500TB of archive data to S3. Over a 1Gbps internet link, this would take 46 days. Using 7 Snowball devices, the transfer completes in 1 week — including shipping time.*

Section 4 · Database Services

Q46	What is Amazon RDS?
A	Amazon Relational Database Service (RDS) is a managed relational database service. AWS handles provisioning, patching, backups, and high availability. Supported engines: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, and Amazon Aurora. You manage the schema and queries.

▯	*Real-World Example: A startup switches from managing their own MySQL server on EC2 (requiring a DBA) to RDS MySQL. AWS now handles nightly backups, OS patching, and failover — the team focuses on building features.*

Q47	What is Amazon Aurora?
A	Amazon Aurora is AWS's cloud-native relational database. Compatible with MySQL and PostgreSQL. Aurora is up to 5x faster than MySQL and 3x faster than PostgreSQL. It automatically replicates 6 copies of data across 3 AZs and supports up to 15 read replicas. Aurora Serverless scales automatically.
▯	*Real-World Example: A fintech company migrates from MySQL to Aurora MySQL-Compatible. Same SQL queries and drivers work, but they see 3x throughput improvement and no performance degradation during failover.*

Q48	What is Amazon DynamoDB?
A	Amazon DynamoDB is a fully managed NoSQL key-value and document database. It delivers single-digit millisecond performance at any scale. DynamoDB automatically scales read/write capacity. It supports DynamoDB Streams, DAX (in-memory caching), and global tables for multi-region replication.
▯	*Real-World Example: Amazon.com's shopping cart uses DynamoDB. During Prime Day, the cart handles millions of writes per second. DynamoDB auto-scales instantly — there's no capacity planning needed.*

Q49	What is Amazon ElastiCache?
A	Amazon ElastiCache is a managed in-memory caching service. Supports Redis and Memcached. Used to cache frequently accessed data, reducing database load and improving application response time from milliseconds to microseconds.
▯	*Real-World Example: A social media app queries the user profile database 10,000 times/second. Adding ElastiCache Redis caches profile data for 5 minutes. 95% of requests hit the cache — database load drops by 95% and response time drops from 50ms to 1ms.*

Q50	What is Amazon Redshift?
A	Amazon Redshift is a fully managed data warehouse service for analytics. It uses columnar storage and parallel query execution, making it optimised for OLAP workloads (complex queries on large datasets). Redshift can query S3 data directly via Redshift Spectrum.

▯	*Real-World Example: An e-commerce company loads 3 years of sales data (500GB) into Redshift. A complex analytics query that would take 10 minutes on MySQL completes in 8 seconds on Redshift — ideal for BI dashboards.*

Q51	What is Amazon Neptune?
A	Amazon Neptune is a fully managed graph database service. It supports Property Graph (Gremlin) and RDF (SPARQL). Used for social networks, fraud detection, knowledge graphs, and recommendation engines where relationships between data points are critical.
▯	*Real-World Example: A fraud detection system uses Neptune to map relationships between bank accounts, devices, and IP addresses. A graph query can identify a fraud ring of 50 connected accounts in milliseconds — impossible with a relational database.*

Q52	What is Amazon DocumentDB?
A	Amazon DocumentDB is a fully managed document database service compatible with MongoDB. It stores data as JSON-like documents. DocumentDB separates compute from storage, replicates data 6 ways across 3 AZs, and supports MongoDB APIs.
▯	*Real-World Example: A team has an existing Node.js app using MongoDB. They migrate to DocumentDB with minimal code changes (same MongoDB driver). AWS now handles backups, replication, and scaling.*

Q53	What is the difference between RDS and DynamoDB?
A	RDS is a relational (SQL) database — structured schema, ACID transactions, complex JOIN queries, vertical scaling. DynamoDB is a NoSQL database — flexible schema, key-value or document model, single-table design, horizontal scaling to any throughput. Choose RDS for complex queries; DynamoDB for high-throughput, low-latency access patterns.
▯	*Real-World Example: Banking transaction records with complex financial queries ▯ use RDS (Aurora). IoT sensor readings at 1 million writes/second ▯ use DynamoDB.*

Section 5 · Networking & Content Delivery

Q54	What is Amazon VPC?
A	Amazon Virtual Private Cloud (VPC) is a logically isolated section of the AWS cloud where you launch AWS resources in a virtual network you define. You control IP address ranges, subnets, route tables, internet gateways, and security. Every AWS account comes with a default VPC.

▯	*Real-World Example: A company creates a VPC 10.0.0.0/16. They create a public subnet (10.0.1.0/24) for web servers and a private subnet (10.0.2.0/24) for databases. The database is not internet-accessible, improving security.*

Q55	What is an Internet Gateway?
A	An Internet Gateway (IGW) is a horizontally scaled, redundant VPC component that enables communication between instances in your VPC and the internet. It performs NAT for instances with public IPs. A VPC can have only one IGW. Public subnets route traffic to the IGW.
▯	*Real-World Example: EC2 instances in a public subnet (web servers) route 0.0.0.0/0 traffic to the IGW, making them internet-accessible. EC2 instances in a private subnet have no IGW route and cannot be reached from the internet directly.*

Q56	What is a NAT Gateway?
A	A NAT (Network Address Translation) Gateway allows EC2 instances in a private subnet to initiate outbound internet connections (e.g., download OS patches) while preventing inbound connections from the internet. The NAT Gateway is placed in a public subnet and routes traffic through the IGW.
▯	*Real-World Example: A database EC2 in a private subnet needs to download security patches. Traffic goes: Database ▯ NAT Gateway (public subnet) ▯ IGW ▯ Internet. No inbound internet traffic can reach the database directly.*

Q57	What is Amazon Route 53?
A	Amazon Route 53 is a scalable, highly available DNS web service. It translates human-readable domain names (www.example.com) to IP addresses. Route 53 supports routing policies: Simple, Weighted, Latency-based, Failover, Geolocation, Multivalue, and Geoproximity.
▯	*Real-World Example: A company has servers in US and India. Route 53 Latency routing sends US users to the US server and Indian users to the Mumbai server — automatically reducing latency for each user group.*

Q58	What is Amazon CloudFront?
A	Amazon CloudFront is a fast Content Delivery Network (CDN) service. It distributes content (static files, videos, APIs) to edge locations worldwide, reducing latency. CloudFront integrates with S3, EC2, ELB, and Lambda@Edge. Provides HTTPS, DDoS protection via AWS Shield.

□ *Real-World Example: Netflix uses CloudFront to serve video streams. A user in London accesses a film stored in S3 (Oregon). CloudFront serves it from the London edge location — reducing buffering from seconds to zero.*

Q59 What is AWS Direct Connect?

A AWS Direct Connect establishes a dedicated private network connection between your on-premises data center and AWS, bypassing the public internet. Provides consistent bandwidth, lower latency, and reduced data transfer costs. Available in 1Gbps and 10Gbps speeds.

□ *Real-World Example: A financial trading firm requires consistent sub-millisecond latency to AWS. They establish a 10Gbps Direct Connect link from their co-location data center to AWS — they cannot tolerate the variable latency of the internet.*

Q60 What is Elastic Load Balancing (ELB)?

A ELB automatically distributes incoming traffic across multiple targets (EC2 instances, containers, IPs, Lambda). Types: Application Load Balancer (ALB) – Layer 7, HTTP/HTTPS; Network Load Balancer (NLB) – Layer 4, TCP/UDP, ultra-high performance; Gateway Load Balancer (GWLB) – third-party virtual appliances.

□ *Real-World Example: A website running on 5 EC2 instances behind an ALB. When EC2-1 becomes unhealthy (health check fails), ALB automatically stops sending it traffic and distributes requests among the remaining 4 healthy instances.*

Q61 What is a VPC Peering connection?

A VPC Peering connects two VPCs (in the same or different AWS accounts/regions) using private IPv4 or IPv6 addresses. Traffic uses the AWS private network, never the internet. Peering is not transitive — if A□B and B□C, A cannot reach C.

□ *Real-World Example: A company has a production VPC and a shared services VPC (with Active Directory). VPC Peering connects them so production EC2 instances can authenticate against AD without traffic traversing the internet.*

Q62 What is AWS VPN?

A AWS VPN provides encrypted tunnels between your on-premises network and AWS VPC over the public internet. Two types: Site-to-Site VPN (connects on-premises network to VPC using IPSec) and Client VPN (allows individual devices to connect to AWS using OpenVPN).

▯	*Real-World Example: A company's employees work from home. AWS Client VPN allows them to connect their laptops securely to the corporate VPC, accessing internal web apps and databases as if they were in the office.*

Q63	What is a Security Group vs NACL?
A	Security Group (SG): Instance-level firewall, stateful (return traffic automatically allowed), only allow rules (no deny). Network ACL (NACL): Subnet-level firewall, stateless (must explicitly allow return traffic), supports both allow and deny rules, rules evaluated in order.
▯	*Real-World Example: Block a specific bad IP (203.0.113.100) from the entire subnet ▯ use NACL with a deny rule. Allow only port 443 inbound to a specific EC2 ▯ use Security Group. Both can be combined for defence in depth.*

Section 6 · Security & Identity

Q64	What is AWS IAM?
A	AWS Identity and Access Management (IAM) is a free service that controls access to AWS resources. You create Users (individual people), Groups (collection of users), Roles (assumed by services/apps), and Policies (JSON documents defining permissions). IAM is global — not region-specific.
▯	*Real-World Example: A new developer joins the company. You create an IAM user, add them to the 'Developers' group. The group has a policy allowing EC2 and S3 read access. The developer inherits those permissions automatically.*

Q65	What is the principle of least privilege?
A	The principle of least privilege means granting users and services only the minimum permissions required to perform their tasks — nothing more. This limits the blast radius if credentials are compromised. Start with zero permissions and add only what is needed.
▯	*Real-World Example: A Lambda function only needs to read from one DynamoDB table. Its IAM role policy allows only dynamodb:GetItem and dynamodb:Query on that specific table ARN. It has no permission to delete, write, or access other tables.*

Q66	What is Multi-Factor Authentication (MFA) in AWS?
A	MFA adds a second layer of security beyond username and password. After entering credentials, users must provide a one-time code from an MFA device (virtual app like Google Authenticator, hardware token, or SMS). AWS strongly recommends enabling MFA on the root account.

▯	*Real-World Example: An attacker steals an IAM user's password. Without MFA, they have full access. With MFA enabled, they also need the 6-digit OTP from the user's phone — making the stolen password alone useless.*

Q67	What is the AWS Root Account?
A	The root account is created when you first sign up for AWS. It has unrestricted access to all AWS services and billing. AWS best practices: never use root for daily tasks, enable MFA on root immediately, create IAM admin users for day-to-day operations, do not create root access keys.
▯	*Real-World Example: A company's root account was compromised because it had no MFA. The attacker launched 100 GPU instances for crypto mining, resulting in a $50,000 bill. This is why AWS mandates MFA on root accounts.*

Q68	What is AWS Organizations?
A	AWS Organizations allows you to centrally manage multiple AWS accounts. You can group accounts into Organizational Units (OUs), apply Service Control Policies (SCPs) to restrict actions, consolidate billing, and enable services across all accounts centrally.
▯	*Real-World Example: A company has 50 AWS accounts (per project/team). Using Organizations with SCPs, they enforce that no account can disable CloudTrail, all resources must be in approved regions only, and all billing is consolidated for volume discounts.*

Q69	What is AWS IAM Role?
A	An IAM Role is an identity with a set of permissions that can be assumed by AWS services, users, or external identities. Unlike users, roles don't have permanent credentials — they issue temporary security tokens. Common use: EC2 instance role to access S3 without storing credentials in code.
▯	*Real-World Example: An EC2 instance running an app needs to write to S3. Instead of hardcoding AWS keys in the code (a security risk), you attach an IAM role with S3 write permissions to the EC2 instance. The app uses the role's temporary credentials automatically.*

Q70	What is AWS CloudTrail?
A	AWS CloudTrail records all API calls and actions taken in your AWS account (who did what, when, from where). Logs are stored in S3. CloudTrail is crucial for security auditing, compliance, and troubleshooting. Enabled by default for 90 days; create trails for longer retention.

▯	*Real-World Example: A production database was accidentally deleted. CloudTrail logs show: IAM user 'john.smith' called DeleteDBInstance at 14:32 UTC from IP 203.0.113.5 using the AWS console. This identifies the responsible person and time for the incident report.*

Q71	What is Amazon GuardDuty?
A	Amazon GuardDuty is an intelligent threat detection service that continuously monitors for malicious activity using machine learning, anomaly detection, and threat intelligence. It analyses CloudTrail, VPC Flow Logs, and DNS logs. No agents to install. Provides findings with severity ratings.
▯	*Real-World Example: GuardDuty detects that EC2 instance i-1234567890 is communicating with a known cryptocurrency mining pool IP at 3AM — a sign of a compromised instance. It generates a High severity finding, which triggers a Lambda to isolate the instance automatically.*

Q72	What is AWS Shield?
A	AWS Shield is a managed DDoS protection service. Shield Standard is automatically enabled at no cost for all AWS customers, protecting against common Layer 3/4 attacks. Shield Advanced (paid, ~$3,000/month) provides enhanced DDoS protection, 24/7 DRT access, and financial protection.
▯	*Real-World Example: A gaming company's website is hit by a 100Gbps UDP flood attack. AWS Shield Standard automatically absorbs and mitigates the attack without any action from the customer — the website remains available.*

Q73	What is AWS WAF?
A	AWS WAF (Web Application Firewall) protects web applications from common web exploits (SQL injection, cross-site scripting). You create rules to allow, block, or monitor web requests. WAF integrates with CloudFront, ALB, API Gateway. Use managed rule groups or create custom rules.
▯	*Real-World Example: A website detects SQL injection attempts. The security team creates a WAF rule: block any request where the query string contains 'DROP TABLE' or '--'. CloudFront applies this rule at the edge, blocking attacks before they reach the application server.*

Q74	What is Amazon Inspector?

A	Amazon Inspector is an automated security assessment service for EC2 instances and container images. It checks for software vulnerabilities (CVEs), unintended network exposure, and deviations from security best practices. Integrates with AWS Security Hub.
▯	*Real-World Example: Inspector scans an EC2 instance and finds it's running Apache 2.4.49 with CVE-2021-41773 (critical path traversal vulnerability). It generates a finding with the CVE details and recommended fix — patch to Apache 2.4.51.*

Q75	What is AWS Secrets Manager?
A	AWS Secrets Manager stores, manages, and automatically rotates sensitive information (database passwords, API keys, OAuth tokens). Applications retrieve secrets via API instead of hardcoding them. Supports automatic rotation for RDS, Redshift, and DocumentDB.
▯	*Real-World Example: A developer hardcoded the RDS password in their app config — a major security risk. They migrate to Secrets Manager. The app calls GetSecretValue at runtime. Secrets Manager rotates the DB password every 30 days automatically — the app always gets the latest value.*

Q76	What is AWS KMS?
A	AWS Key Management Service (KMS) is a managed service for creating and managing encryption keys. KMS integrates with 100+ AWS services. You can encrypt S3 objects, EBS volumes, RDS databases using KMS keys (CMKs). Provides audit logs via CloudTrail.
▯	*Real-World Example: A healthcare company stores patient records in S3 with KMS encryption (SSE-KMS). Even if an attacker gains S3 access, they cannot read the data without access to the KMS key. Key usage is logged in CloudTrail for compliance.*

Q77	What is Amazon Cognito?
A	Amazon Cognito provides user authentication and authorisation for web and mobile apps. User Pools manage user registration and sign-in (with social login via Google/Facebook). Identity Pools grant AWS credentials to authenticated users. Supports MFA, email verification.
▯	*Real-World Example: A mobile app (fitness tracker) uses Cognito User Pool for sign-up/sign-in. After login, Cognito Identity Pool gives the user temporary AWS credentials with permission to access only their own S3 folder for workout data.*

Q78	What is the AWS Trusted Advisor?

A	AWS Trusted Advisor is an online tool that provides real-time guidance across 5 categories: Cost Optimization, Performance, Security, Fault Tolerance, and Service Limits. Basic checks are free; full access requires Business or Enterprise Support plan.
▯	*Real-World Example: Trusted Advisor alerts a company: 'You have 3 S3 buckets with public read access' (Security) and 'Your EC2 instances have been idle for 14 days' (Cost Optimization). Acting on these findings saves money and improves security.*

Section 7 · Management & Governance

Q79	What is AWS CloudFormation?
A	AWS CloudFormation is an Infrastructure as Code (IaC) service. You define AWS resources in JSON or YAML templates. CloudFormation creates, updates, and deletes resources as a 'stack'. Enables repeatable, version-controlled infrastructure deployments.
▯	*Real-World Example: A DevOps team writes a CloudFormation template (YAML) that creates a VPC, 2 subnets, an ALB, an Auto Scaling group, and an RDS instance. They deploy identical stacks in dev, staging, and prod environments — each takes 5 minutes, always identical.*

Q80	What is AWS Config?
A	AWS Config continuously monitors and records the configurations of AWS resources. It evaluates resource configurations against desired rules (e.g., 'all S3 buckets must have encryption enabled'). Provides configuration history and change notifications.
▯	*Real-World Example: An auditor asks for proof that all EC2 instances used encryption in 2023. AWS Config shows the full configuration history of every EC2 instance — when it was created, what settings it had, and every change made — perfect for compliance audits.*

Q81	What is Amazon CloudWatch?
A	Amazon CloudWatch is a monitoring service for AWS resources and applications. It collects metrics (CPU, memory, network), logs (CloudWatch Logs), and events. You create dashboards, alarms (notify via SNS when CPU > 80%), and automated actions. CloudWatch agent collects OS-level metrics.
▯	*Real-World Example: A company sets a CloudWatch alarm: if EC2 CPU > 90% for 5 minutes, send an SNS email to ops@company.com and trigger Auto Scaling to add 2 instances. This is proactive monitoring — problems are resolved before users notice.*

Q82	What is AWS Systems Manager?
A	AWS Systems Manager provides a unified interface for viewing and controlling AWS infrastructure. Key features: Session Manager (secure shell access to EC2 without SSH/bastion), Patch Manager (automated OS patching), Parameter Store (configuration data), Run Command (execute scripts at scale).
▯	*Real-World Example: An ops team manages 500 EC2 instances. Instead of SSH-ing into each one, they use Systems Manager Run Command to execute 'yum update -y' on all 500 instances simultaneously — patched in minutes, with full audit logs.*

Q83	What is AWS Cost Explorer?
A	AWS Cost Explorer is a tool for visualising, understanding, and managing AWS costs. View spending by service, account, tag, or region. Identify cost trends, set budgets, and right-size recommendations. Provides Reserved Instance and Savings Plan recommendations.
▯	*Real-World Example: A CTO reviews Cost Explorer and notices EC2 costs spike every Monday morning. Drill-down shows a batch processing job runs 200 on-demand instances. They switch to Spot Instances, saving 70% — $8,000/month.*

Q84	What is AWS Budgets?
A	AWS Budgets allows you to set custom cost and usage budgets and receive alerts when you exceed (or are forecasted to exceed) thresholds. Types: Cost Budget, Usage Budget, Reservation Budget, Savings Plan Budget. Alerts can trigger SNS notifications or actions (stop EC2 instances).
▯	*Real-World Example: A startup sets a $500/month AWS Budget. At 80% threshold ($400), AWS emails the team lead. At 100% ($500), an automated action stops all non-production EC2 instances — preventing runaway costs from a developer's forgotten test environment.*

Q85	What is AWS Control Tower?
A	AWS Control Tower automates the setup and governance of a multi-account AWS environment. It sets up a landing zone with pre-configured security and compliance guardrails, using AWS Organizations, CloudTrail, Config, and IAM Identity Center.
▯	*Real-World Example: A company wants to create 50 AWS accounts for 50 teams. Without Control Tower, each account would need manual security setup. With Control Tower, all 50 accounts are provisioned with consistent guardrails (MFA required, CloudTrail enabled, no public S3 buckets) automatically.*

Q86 What is AWS Service Catalog?

A AWS Service Catalog allows organisations to create and manage catalogues of approved IT services that users can deploy. Ensures governance and compliance — developers deploy from pre-approved, standardised templates instead of creating resources ad-hoc.

 Real-World Example: A bank's cloud team pre-approves 10 types of infrastructure (e.g., 'Standard Web Application', 'Secure Database'). Developers pick from the Service Catalog. They can't create non-compliant resources — every deployment meets security standards automatically.

Q87 What is Amazon EventBridge?

A Amazon EventBridge is a serverless event bus. It routes events from AWS services, custom applications, and SaaS (Salesforce, Zendesk) to targets (Lambda, SQS, SNS). Enables event-driven architectures. Formerly called CloudWatch Events.

 Real-World Example: When a new file is uploaded to S3 (PutObject event), EventBridge routes this event to a Lambda function that processes the file and to an SQS queue that notifies downstream services — all without any polling or custom code connecting the services.

Section 8 · Application Integration

Q88 What is Amazon SQS?

A Amazon Simple Queue Service (SQS) is a fully managed message queuing service. It decouples application components — producer sends messages to a queue; consumer retrieves and processes them independently. Two types: Standard Queue (at-least-once delivery, best-effort ordering) and FIFO Queue (exactly-once, strict order).

 Real-World Example: An order processing system: web app puts order messages into SQS. 5 backend workers each pull messages from the queue to process payments. If one worker crashes, its message goes back to the queue and another worker processes it — no orders lost.

Q89 What is Amazon SNS?

A	Amazon Simple Notification Service (SNS) is a publish-subscribe messaging service. Publishers send messages to a Topic; all subscribers receive them simultaneously (fan-out). Supports: SQS queues, Lambda, HTTP endpoints, email, SMS. Used for notifications and event-driven fan-out.
▯	*Real-World Example: An e-commerce app publishes an 'Order Placed' event to an SNS Topic. 3 subscribers receive it simultaneously: Lambda (sends confirmation email), SQS (queues order for warehouse), another SQS (notifies accounting system). One publish ▯ 3 actions.*

Q90	What is Amazon API Gateway?
A	Amazon API Gateway is a fully managed service for creating, publishing, securing, and monitoring REST, HTTP, and WebSocket APIs. It acts as a 'front door' for applications. Integrates with Lambda (serverless APIs), EC2, and any HTTP endpoint. Handles auth, throttling, caching.
▯	*Real-World Example: A mobile app calls https://api.company.com/products. API Gateway receives the request, validates the JWT token, throttles to 1000 req/sec, and invokes a Lambda function that queries DynamoDB. The mobile app never directly touches Lambda or DynamoDB.*

Q91	What is Amazon SES?
A	Amazon Simple Email Service (SES) is a cloud-based email sending service. Used for transactional emails (order confirmations, password resets), marketing emails, and bulk sending. Provides high deliverability, with DKIM and SPF support. Also supports receiving email.
▯	*Real-World Example: An e-commerce platform sends 5 million order confirmation emails/month. Using SES at $0.10/1000 emails = $500/month. Their previous SendGrid plan was $2,000/month — SES saves 75% on email costs.*

Q92	What is AWS Step Functions?
A	AWS Step Functions is a serverless workflow orchestration service. You define workflows as state machines (JSON/YAML). Coordinates multiple Lambda functions, ECS tasks, and AWS services into multi-step workflows with error handling, retries, and parallel execution.
▯	*Real-World Example: An insurance claim workflow: 1) Receive claim (Lambda) ▯ 2) Fraud check (Lambda, parallel with) ▯ 3) Document OCR (Textract) ▯ 4) If fraud detected: manual review; else: auto-approve ▯ 5) Send notification (SNS). Step Functions orchestrates this without custom glue code.*

Section 9 · Developer Tools & DevOps

Q93 What is AWS CodePipeline?

A AWS CodePipeline is a fully managed CI/CD service that automates the release process. A pipeline has stages: Source (CodeCommit/GitHub) □ Build (CodeBuild) □ Test □ Deploy (CodeDeploy/Elastic Beanstalk/ECS). Every code commit automatically triggers the pipeline.

□ *Real-World Example: A developer pushes code to GitHub. CodePipeline triggers: CodeBuild runs unit tests and builds a Docker image □ pushes to ECR □ CodeDeploy performs a blue/green deployment to ECS. The entire process takes 8 minutes, fully automated.*

Q94 What is AWS CodeBuild?

A AWS CodeBuild is a fully managed CI service that compiles source code, runs tests, and produces deployment artifacts. You provide a buildspec.yml file defining build commands. CodeBuild scales automatically — no build servers to manage.

□ *Real-World Example: A buildspec.yml runs: npm install □ npm test □ npm run build □ docker build □ docker push to ECR. CodeBuild executes this in an isolated container every time code is committed, ensuring consistent builds.*

Q95 What is AWS CodeDeploy?

A AWS CodeDeploy automates application deployments to EC2 instances, on-premises servers, Lambda functions, and ECS services. Supports deployment strategies: In-place (rolling update), Blue/Green (new environment □ traffic shift □ terminate old), Canary (gradually shift traffic).

□ *Real-World Example: A company deploys a new API version using CodeDeploy Blue/Green: new EC2 instances (green) are launched with the new code. CodeDeploy shifts 10% of traffic to green, monitors for errors for 10 minutes, then shifts 100% — zero-downtime deployment.*

Q96 What is AWS CodeCommit?

A AWS CodeCommit is a fully managed source control service that hosts private Git repositories. It integrates with IAM for access control, supports pull requests and code reviews, and triggers CodePipeline automatically on commits. Data is encrypted at rest and in transit.

Q97 What is AWS Cloud9?

A	AWS Cloud9 is a cloud-based integrated development environment (IDE) accessed through a web browser. It provides a code editor, debugger, and terminal with direct AWS service access. Eliminates local development environment setup — useful for teams and educational settings.

Q98	What is AWS X-Ray?
A	AWS X-Ray helps developers analyse and debug distributed applications (microservices). It traces requests end-to-end across Lambda, EC2, ECS, API Gateway, and DynamoDB. Provides a service map showing request flow, latency, and error rates between services.
▯	*Real-World Example: A user reports that their checkout page takes 8 seconds. X-Ray traces show: API Gateway (50ms) ▯ Lambda (100ms) ▯ DynamoDB (7,500ms). The bottleneck is DynamoDB — an engineer adds DAX caching and checkout drops to 300ms.*

Section 10 · Analytics

Q99	What is Amazon Athena?
A	Amazon Athena is a serverless, interactive query service that uses standard SQL to analyse data directly in S3. No infrastructure to manage. You pay per query ($5/TB scanned). Supports CSV, JSON, Parquet, ORC formats. Integrates with AWS Glue Data Catalog.
▯	*Real-World Example: A company stores 10TB of CloudFront access logs in S3 as gzipped CSV files. Using Athena, an analyst writes: SELECT uri, COUNT(*) FROM logs WHERE date='2024-01-15' GROUP BY uri ORDER BY 2 DESC. Results in 30 seconds — no data warehouse needed.*

Q100	What is Amazon Kinesis?
A	Amazon Kinesis is a platform for real-time streaming data. Components: Kinesis Data Streams (collect and process streaming data), Kinesis Data Firehose (load streams to S3/Redshift/OpenSearch), Kinesis Data Analytics (SQL on streams in real-time), Kinesis Video Streams (video ingestion).
▯	*Real-World Example: A ride-sharing app ingests 1 million GPS location updates per second from drivers. Kinesis Data Streams collects them; Kinesis Data Analytics runs SQL to detect drivers near unserved passengers in real-time; Firehose delivers raw data to S3 for later analysis.*

Q101	What is AWS Glue?

A	AWS Glue is a fully managed ETL (Extract, Transform, Load) service. It discovers data (Glue Crawler populates the Data Catalog), transforms it (PySpark/Python scripts in Glue Jobs), and loads it to destinations (S3, Redshift, RDS). Serverless — no infrastructure to manage.
▯	*Real-World Example: A company has sales data in CSV (S3), user data in MySQL (RDS), and log data in JSON (S3). Glue crawls all sources, creates a unified Data Catalog, then a Glue ETL job joins and transforms the data and loads it to Redshift for analytics.*

Q102	What is Amazon QuickSight?
A	Amazon QuickSight is a cloud-native business intelligence (BI) and data visualization service. It connects to S3, Redshift, RDS, Athena, and third-party data sources. Create interactive dashboards and reports. Uses SPICE (Super-fast, Parallel, In-memory Calculation Engine) for fast queries.
▯	*Real-World Example: A retail company's finance team uses QuickSight to create a live sales dashboard. Data from Redshift is visualised as bar charts, heat maps, and trend lines. The CEO views it on their phone — no SQL knowledge required.*

Section 11 · Machine Learning & AI

Q103	What is Amazon SageMaker?
A	Amazon SageMaker is a fully managed ML platform for building, training, and deploying ML models. Features: SageMaker Studio (IDE), built-in algorithms, managed Jupyter notebooks, training at scale, one-click model deployment, A/B testing endpoints.
▯	*Real-World Example: A data scientist builds a customer churn prediction model. They use SageMaker to: prepare data (Data Wrangler) ▯ train XGBoost model on 100 GPU instances ▯ evaluate accuracy ▯ deploy to a SageMaker endpoint. The whole ML lifecycle in one platform.*

Q104	What is Amazon Rekognition?
A	Amazon Rekognition is a computer vision service using deep learning to analyse images and videos. Features: object and scene detection, facial analysis and recognition, text in images, content moderation, celebrity recognition, activity detection.
▯	*Real-World Example: A media platform uses Rekognition to automatically detect and blur faces of children in user-uploaded photos (content moderation). Processing 1 million images takes minutes and costs fractions of a cent per image — no ML expertise needed.*

Q105	What is Amazon Textract?
A	Amazon Textract extracts text and structured data from scanned documents. Unlike basic OCR, Textract understands document structure — it can extract tables, forms (key-value pairs), and checkboxes from PDFs and images.
▯	*Real-World Example: An insurance company receives 10,000 claim forms daily as scanned PDFs. Textract extracts the claim number, policy holder, date, and medical data fields automatically. This replaces a team of 20 data entry employees.*

Q106	What is Amazon Polly?
A	Amazon Polly converts text to realistic speech using deep learning. Supports 60+ voices in 30+ languages, including neural TTS voices that sound very human-like. SSML markup for pronunciation control. Output as MP3/OGG streaming audio.
▯	*Real-World Example: A language learning app uses Polly to generate pronunciation examples. A user types 'Bonjour, comment allez-vous?' and Polly instantly generates a natural-sounding French audio clip — no voice recording studio needed.*

Q107	What is Amazon Lex?
A	Amazon Lex is a service for building conversational interfaces (chatbots) using voice and text. Uses the same deep learning technology as Amazon Alexa. Supports intents, slots, and fulfilment via Lambda. Integrates with Connect, Messenger, Slack, Twilio.
▯	*Real-World Example: A bank builds a customer service chatbot using Lex. A customer says 'What's my account balance?' Lex identifies the intent (CheckBalance), asks for account number (slot), calls Lambda to query the database, and responds with the balance in natural language.*

Q108	What is Amazon Translate?
A	Amazon Translate is a neural machine translation service for translating text between languages. Supports 75+ languages. Real-time and batch translation. Integrates with S3, Comprehend, and other AWS services. Used for multilingual apps and content localisation.

Q109	What is Amazon Comprehend?

A	Amazon Comprehend is a natural language processing (NLP) service. It identifies language, extracts key phrases, entities (people, places, organisations), sentiment (positive/negative/neutral), and topics from text. Comprehend Medical extracts medical information from clinical text.
▯	*Real-World Example: A company analyses 100,000 customer reviews from Amazon. Comprehend identifies that 73% of 1-star reviews mention 'delivery' (entity) with negative sentiment — revealing a logistics problem that customer service metrics missed.*

Section 12 · Billing & Pricing

Q110	How does AWS pricing work?
A	AWS uses a pay-as-you-go model. You pay only for what you use with no upfront costs (in on-demand model). Three pricing fundamentals: 1. Compute (per hour/second). 2. Storage (per GB/month). 3. Data Transfer (outbound data is charged; inbound is free). Prices vary by region.
▯	*Real-World Example: An EC2 t3.micro running 24/7 for 30 days: 720 hours × \$0.0104/hour = \$7.49/month. An S3 bucket with 100GB: 100 × \$0.023/GB = \$2.30/month. Data transfer out: 10GB × \$0.09/GB = \$0.90. Total: ~\$10.69/month.*

Q111	What is the AWS Free Tier?
A	AWS Free Tier provides limited free usage of many services for 12 months after account creation, plus some services always free. Examples: EC2 t2.micro (750 hours/month for 12 months), S3 (5GB for 12 months), Lambda (1M free requests/month always free), DynamoDB (25GB always free).
▯	*Real-World Example: A student learning AWS creates a Free Tier account. They run a t2.micro EC2 instance for a web project all month without charges (750 hours free). Their Lambda function handles 500K requests (under 1M always-free limit) — total bill: \$0.*

Q112	What are Reserved Instances (RI)?
A	Reserved Instances provide a significant discount (up to 72%) compared to On-Demand pricing in exchange for a commitment to use a specific instance type in a specific region. Terms: 1-year or 3-year. Payment options: All Upfront (biggest discount), Partial Upfront, No Upfront.
▯	*Real-World Example: A company runs m5.xlarge EC2 24/7 in us-east-1. On-Demand: \$0.192/hr × 8760 hrs = \$1,682/year. 1-year All-Upfront RI: \$1,022. Saving: \$660/year (39% discount) — justified because the instance is always running.*

Q113	What are Savings Plans?
A	Savings Plans are a flexible pricing model offering up to 72% discount. Two types: Compute Savings Plans (most flexible — applies to EC2, Lambda, Fargate across any region/instance family) and EC2 Instance Savings Plans (higher discount but locked to a specific instance family in a region).
◻	*Real-World Example: A company commits to spending $100/hour on compute for 1 year (Compute Savings Plans). This applies automatically to EC2, Lambda, and Fargate usage — even if they change instance types or move regions. No need to predict exact instance needs.*

Q114	What is AWS Total Cost of Ownership (TCO)?
A	TCO is the total cost of IT infrastructure including hardware, software, facility, power, cooling, IT staff, and maintenance. The AWS TCO Calculator helps compare on-premises costs to AWS costs. Typically, companies find 30-60% cost savings moving to AWS.
◻	*Real-World Example: On-premises 3-year TCO: $1.2M (servers $500K, facility $300K, power $150K, staff $250K). AWS equivalent: $420K (EC2 Reserved Instances + staff reductions). AWS TCO saving: $780K over 3 years.*

Q115	What is AWS Pricing Calculator?
A	The AWS Pricing Calculator (calculator.aws) estimates the cost of AWS services before you use them. You configure the services you plan to use, and it provides a monthly and annual cost estimate. Useful for budgeting and comparing architecture options.
◻	*Real-World Example: Before launching a new product, a CTO uses the Pricing Calculator to estimate costs: 2 EC2 t3.large + 1 RDS db.t3.medium + 1TB S3 + CloudFront = $347/month estimated. Helps make build-vs-buy decision.*

Q116	What is AWS Consolidated Billing?
A	Consolidated Billing in AWS Organizations combines the bills from all member accounts into a single master/management account. Benefits: one invoice, volume discounts (all accounts' usage is combined to reach higher discount tiers), Reserved Instance/Savings Plan sharing across accounts.
◻	*Real-World Example: A company has 10 AWS accounts each using 500GB S3. Combined, that's 5TB — qualifying for S3 volume pricing tier (cheaper per GB). Consolidated billing automatically applies the discount across all accounts.*

Q117	What is the difference between On-Demand, Reserved, and Spot Instances?
A	On-Demand: Pay by the second, no commitment, highest price, use for unpredictable workloads. Reserved: 1 or 3-year commitment, up to 72% discount, use for steady-state workloads. Spot: Bid for unused capacity, up to 90% discount, can be interrupted with 2-min notice, use for fault-tolerant batch jobs.
▯	*Real-World Example: A media company: web servers (always running) ▯ Reserved Instances. Video encoding (can be interrupted) ▯ Spot Instances (saves 80%). Traffic surge during breaking news (unpredictable) ▯ On-Demand Auto Scaling. All three types in the same architecture.*

Q118	What is AWS Cost Allocation Tags?
A	Cost Allocation Tags are key-value pairs attached to AWS resources to categorise costs. Activated in the Billing Console, they appear in Cost Explorer and billing reports. Two types: AWS-generated (aws:createdBy) and user-defined (Project, Environment, Team, CostCenter).
▯	*Real-World Example: A company tags all resources: Environment=Production/Development, Project=Checkout/Search, Team=Backend/Frontend. Cost Explorer shows Project=Checkout spends $15K/month and Team=Backend spends $22K/month — enabling chargeback to individual project budgets.*

Section 13 · Migration & Transfer

Q119	What is the AWS Cloud Adoption Framework (CAF)?
A	The AWS CAF provides guidance for cloud migration and transformation. It has 6 perspectives: Business (strategy alignment), People (org change management), Governance (risk, compliance), Platform (cloud architecture), Security (threat protection), Operations (day-to-day management).
▯	*Real-World Example: A traditional bank uses the CAF Business Perspective to build a business case (ROI, risk reduction) for cloud migration. The People Perspective identifies that 200 IT staff need AWS training. The Platform Perspective designs the target hybrid cloud architecture.*

Q120	What are the 7 Rs of cloud migration?

A	1. Retire – turn off unused apps. 2. Retain – keep on-premises (for now). 3. Rehost (lift-and-shift) – move as-is to EC2. 4. Relocate – move to VMware Cloud on AWS. 5. Replatform – minor optimisations (MySQL RDS). 6. Repurchase – switch to SaaS. 7. Refactor/Re-architect – redesign as cloud-native.
	Real-World Example: Company migration: CRM Repurchase (switch to Salesforce SaaS). Legacy mainframe Retain (too risky to migrate now). 50 Java apps Rehost to EC2 (fast migration). MySQL Replatform to RDS Aurora (minimal changes, big managed service benefit).

Q121	What is AWS DMS?
A	AWS Database Migration Service (DMS) migrates databases to AWS with minimal downtime. Supports homogeneous migrations (MySQL RDS MySQL) and heterogeneous (Oracle Aurora PostgreSQL, using Schema Conversion Tool). DMS replicates ongoing changes during cutover.
	Real-World Example: A company needs to migrate a 2TB Oracle database to Aurora PostgreSQL with zero downtime. DMS performs an initial full load (takes 12 hours), then continuously replicates changes (CDC). At cutover, they switch the app connection string — only seconds of downtime.

Q122	What is AWS Snow Family?
A	The Snow Family provides physical devices for large data transfers and edge computing: Snowcone (8TB, smallest, battery-powered), Snowball Edge (80TB, compute + storage), Snowmobile (100PB, shipping container). Used when internet bandwidth is insufficient for migration.
	Real-World Example: A film production company has 3PB of 8K raw footage to migrate to S3. Uploading over 1Gbps internet would take 277 days. AWS sends a Snowmobile truck to their facility. They load data in 10 days and AWS uploads to S3 — migration complete in 2 weeks.

Q123	What is AWS Application Migration Service?
A	AWS Application Migration Service (MGN) — formerly CloudEndure Migration — is the primary AWS service for lift-and-shift migrations. It continuously replicates source servers to AWS and provides non-disruptive testing. Used for large-scale server migrations to EC2.

▯	*Real-World Example: A company has 300 on-premises servers to migrate. MGN agent is installed on each. It continuously replicates data to AWS in the background. During maintenance windows, they perform test cutover, verify apps work, then do final cutover — minimal downtime per server.*

Section 14 · Support Plans & Resources

Q124	What are the AWS Support Plans?
A	AWS has 5 support plans: 1. Basic (free) – forums, documentation, Trusted Advisor 7 checks. 2. Developer ($29/month) – one contact, business hours email, general guidance. 3. Business ($100+/month) – 24/7 phone/chat, full Trusted Advisor, <1 hour critical response. 4. Enterprise On-Ramp ($5,500+/month) – pool of TAMs. 5. Enterprise ($15,000+/month) – dedicated TAM, <15 min critical response.
▯	*Real-World Example: A bank with a mission-critical trading platform uses Enterprise Support. When a production database goes down at 2AM Friday, they call AWS and get a response in 15 minutes from a dedicated Technical Account Manager who helps resolve the issue.*

Q125	What is a Technical Account Manager (TAM)?
A	A TAM is a dedicated AWS expert assigned to Enterprise and Enterprise On-Ramp support customers. They provide proactive guidance, architectural reviews, escalation management, and regular business reviews. Available for Enterprise plans only.
▯	*Real-World Example: A company's TAM schedules a quarterly Well-Architected Review of their e-commerce platform. The TAM identifies that they lack multi-AZ RDS setup — a single point of failure. They help design and implement the fix before it causes an outage.*

Q126	What is the AWS Partner Network (APN)?
A	The AWS Partner Network is a global community of AWS-trained and certified companies. Technology Partners build AWS-integrated products. Consulting Partners provide advisory and implementation services. Partners are tiered: Registered, Select, Advanced, Premier.
▯	*Real-World Example: A company hiring for a complex AWS migration engages an AWS Premier Consulting Partner (e.g., Accenture or Deloitte). The partner's hundreds of AWS-certified architects design and execute the migration — validated by AWS as expert-level.*

Q127	What is the AWS Marketplace?
A	AWS Marketplace is a digital catalogue with thousands of software listings from independent vendors. You can find, test, buy, and deploy software (AMIs, SaaS, containers) with billing added to your AWS bill. Categories: security, ML, networking, storage, DevOps.
▯	*Real-World Example: A company needs a next-gen firewall. Instead of procuring hardware, they find Palo Alto Networks VM-Series Firewall on AWS Marketplace. One-click deployment to their VPC. License cost added to their monthly AWS bill — no separate procurement process.*

Q128	What is the AWS Well-Architected Tool?
A	The AWS Well-Architected Tool is a free service in the AWS Console that helps you review architectures against the 6 pillars of the Well-Architected Framework. You answer questions about your workload and receive a report with findings and improvement recommendations.
▯	*Real-World Example: A startup uses the tool to review their new SaaS application. It identifies 3 High-Risk Issues: no automated backups, no multi-AZ database, and hardcoded credentials in Lambda. The team addresses these before launch, avoiding potential disasters.*

Q129	What is AWS IQ?
A	AWS IQ connects customers with AWS Certified freelancers and consulting firms for on-demand AWS project help. Customers post a project, experts apply, and work is performed remotely. Payment is processed securely through AWS — charged to the AWS account.
▯	*Real-World Example: A company needs a one-week engagement to migrate a legacy app to serverless. They post on AWS IQ, receive proposals from 5 AWS-certified experts, select one at $200/hour, and the migration is completed in 5 days — no long-term consulting contract required.*

Q130	What is AWS re:Post?
A	AWS re:Post (formerly AWS Forums) is a community Q&A; service where customers ask technical questions and get answers from AWS experts, partners, and the community. It gamifies participation with reputation points and badges. Free for all AWS customers.

Q131	What is the AWS Personal Health Dashboard?
A	The AWS Personal Health Dashboard (now part of AWS Health) provides alerts and guidance when AWS is experiencing events that may affect your specific resources. Unlike the general Service Health Dashboard, it shows issues relevant only to YOUR account and resources.
▯	*Real-World Example: AWS has a networking issue in us-east-1. The AWS Health Dashboard shows: 'Your EC2 instances i-12345 and i-67890 in us-east-1a may be affected.' You receive a notification 10 minutes before the issue begins — giving time to fail over to us-east-1b.*

Section 15 · Cloud Architecture Patterns

Q132	What is a microservices architecture?
A	Microservices is an architectural style that structures an application as a collection of small, independently deployable services, each responsible for a specific business capability. Each service runs in its own process, communicates via APIs, and can be deployed, scaled, and updated independently.
▯	*Real-World Example: Netflix deconstructed their monolithic DVD rental app into 700+ microservices: User Authentication, Recommendations, Streaming, Billing, etc. Each team owns and deploys their microservice independently — a new recommendation algorithm ships without touching the streaming code.*

Q133	What is a monolithic vs microservices architecture?
A	A monolith packages all application components (UI, business logic, database layer) into a single deployable unit. All components must be deployed together. Microservices splits the app into independent services. Monolith is simpler to start but harder to scale; microservices are complex but independently scalable and deployable.
▯	*Real-World Example: An e-commerce monolith: deploying a bug fix in payments requires redeploying the entire app (including search, user profiles, etc.) — high risk, slow deployment. With microservices, the payment service deploys independently in 5 minutes with no risk to other services.*

Q134	What is an event-driven architecture?

A	Event-driven architecture (EDA) uses events (state changes) as triggers for communication between services. A producer emits an event; consumers react independently. Decouples services — producers don't know about consumers. AWS services: EventBridge, SNS, SQS, Kinesis.
▯	*Real-World Example: An order placed event triggers: Inventory service (reserves stock), Email service (sends confirmation), Analytics service (updates dashboard), Fraud service (checks for fraud) — all simultaneously, all independently, none blocking the others.*

Q135	What is the Strangler Fig pattern?
A	The Strangler Fig pattern gradually migrates a monolith to microservices. New features are built as microservices. Over time, functionality is extracted from the monolith piece by piece. Eventually, the monolith is 'strangled' — replaced entirely without a big-bang rewrite.
▯	*Real-World Example: A company's monolithic e-commerce app: First, extract the product search feature as a microservice (3 months). Then extract user authentication (2 months). Then payments (4 months). After 2 years, the monolith is gone — replaced incrementally with zero service disruption.*

Q136	What is a serverless architecture?
A	Serverless architecture uses managed services where you don't provision or manage servers. Compute: Lambda. API: API Gateway. Database: DynamoDB. Storage: S3. Queue: SQS. Auth: Cognito. Notification: SNS. The cloud provider manages all infrastructure — you focus purely on business logic.
▯	*Real-World Example: A startup builds a complete backend using: API Gateway (REST API) ▯ Lambda (business logic) ▯ DynamoDB (database) ▯ S3 (file storage) ▯ Cognito (auth) ▯ SES (email). Monthly cost with low traffic: ~$5. The team of 2 developers manages zero servers.*

Q137	What is circuit breaker pattern?
A	The Circuit Breaker pattern prevents cascading failures in distributed systems. When a downstream service fails repeatedly, the circuit 'opens' (stops sending requests) for a cooldown period, returns a fallback response, then periodically tests if the service has recovered. AWS: implemented with Lambda error handling and Step Functions.

□	*Real-World Example: Payment service calls Fraud Detection API. Fraud API starts timing out. Without circuit breaker: all payment requests hang for 30 seconds, exhausting threads, crashing the payment service. With circuit breaker: after 5 failures, circuit opens, payments continue with 'fraud check pending' status — service remains functional.*

Q138	What is the CQRS pattern?
A	CQRS (Command Query Responsibility Segregation) separates read (Query) and write (Command) operations into different models. Write operations go to a command database (optimised for writes); reads go to a query database (optimised for reads, often denormalised). Improves performance and scalability.
□	*Real-World Example: An e-commerce app writes orders to an Aurora MySQL (command store). Separately, DynamoDB stores a denormalised order history per user (query store). Customer dashboard reads from DynamoDB (<5ms, highly cached). Order processing writes to Aurora (full ACID transactions). Neither workload impacts the other.*

Q139	What is AWS Well-Architected Framework Performance Efficiency pillar?
A	The Performance Efficiency pillar focuses on using computing resources efficiently as demand changes. Best practices: choose the right EC2 instance types, use serverless where possible, leverage caching (ElastiCache), use CDN (CloudFront), monitor performance with CloudWatch, and consider database-specific optimisation (Aurora vs DynamoDB).
□	*Real-World Example: A company discovers from CloudWatch that their Lambda function runs 3,000ms each invocation — but only does work for 200ms (2,800ms is cold start). They enable Lambda Provisioned Concurrency — cold starts eliminated, latency drops to 210ms. Performance Efficiency improvement with one configuration change.*

Q140	What is multi-tier architecture?
A	Multi-tier (n-tier) architecture separates application concerns into distinct layers. Classic 3-tier: Presentation Tier (web/mobile client or ALB+EC2 web servers), Application Tier (EC2/Lambda business logic in private subnet), Data Tier (RDS/DynamoDB in private subnet). Each tier scales independently.
□	*Real-World Example: Architecture: Users □ CloudFront □ ALB (public subnet) □ EC2 Auto Scaling (app servers, private subnet) □ RDS Aurora (database, private subnet). Each tier in its own subnet and security group. Web tier scales for traffic; database tier scales for data volume. Separation of concerns and defence in depth.*

Q141	What is loosely coupled vs tightly coupled architecture?
A	Tightly coupled: services directly call each other synchronously; if one fails, the caller fails. Loosely coupled: services communicate via queues, events, or APIs; components are independent; failures are isolated. AWS promotes loosely coupled architectures using SQS, SNS, EventBridge.
▯	*Real-World Example: Tightly coupled: Order service calls Inventory service directly (REST). If Inventory is down, orders fail. Loosely coupled: Order service sends message to SQS. Inventory service pulls from SQS when available. Orders queue up during Inventory downtime — no orders lost, no cascading failure.*

Q142	What is immutable infrastructure?
A	Immutable infrastructure means servers are never modified after deployment — instead, new servers are built from scratch and old ones replaced. No patching in-place, no configuration drift. Achieved with AMIs (bake in configuration), Auto Scaling, and CloudFormation rolling updates.
▯	*Real-World Example: Traditional (mutable): SSH into production EC2, run apt-get update, install new library. Problem: servers become 'snowflakes' — each slightly different. Immutable: Build new AMI with updated library, create new Auto Scaling group, route traffic to new ASG, terminate old ASG. Every server is identical and reproducible.*

Section 16 · AWS Cost Management Deep Dive

Q143	What is AWS Cost and Usage Report (CUR)?
A	The AWS Cost and Usage Report is the most detailed billing dataset available. It provides hourly or daily line items for all AWS services used, including pricing, usage, tags, and Reserved Instance allocation. Delivered to S3 as CSV or Parquet files. Queryable with Athena for deep cost analysis.
▯	*Real-World Example: A FinOps team uses CUR + Athena to query: SELECT service, SUM(unblended_cost) FROM cur WHERE month='2024-01' GROUP BY service ORDER BY 2 DESC. They discover EC2 ($45K), RDS ($22K), Data Transfer ($15K) are top costs — prioritising optimisation efforts.*

Q144	What are AWS Savings Plans vs Reserved Instances differences?

A	Reserved Instances (RIs): commit to a specific instance type in a specific region (e.g., m5.xlarge in us-east-1). Higher discount (up to 72%) but less flexible. Savings Plans: commit to a spend amount ($/hour). Compute Savings Plans apply across EC2, Lambda, Fargate, any region, any instance family. More flexible, same max discount.
▯	*Real-World Example: Company A knows they'll always run m5.xlarge in us-east-1 ▯ EC2 Reserved Instance (highest discount). Company B uses many instance types and plans to migrate regions ▯ Compute Savings Plans (flexibility without losing discount). Both save ~65% vs on-demand.*

Q145	What is EC2 Spot Instance interruption?
A	Spot Instances can be reclaimed by AWS with 2 minutes notice when EC2 capacity is needed for On-Demand customers. AWS sends an interruption notification via EC2 instance metadata and EventBridge. Applications must handle interruptions gracefully: checkpoint work, drain requests, save state.
▯	*Real-World Example: A video transcoding job on Spot Instances gets interrupted after 45 minutes. The application detects the 2-minute warning (via EventBridge), saves progress to S3 checkpoint, terminates gracefully. When a new Spot Instance launches, it resumes from the checkpoint — no work is lost.*

Q146	What is AWS Compute Optimizer?
A	AWS Compute Optimizer uses ML to analyse EC2 instances, Auto Scaling groups, EBS volumes, and Lambda functions and recommends optimal configurations. It identifies over-provisioned resources (waste) and under-provisioned resources (performance issues). Reduces costs by up to 25%.
▯	*Real-World Example: Compute Optimizer analyses 50 EC2 instances. It recommends: 20 instances are over-provisioned (average CPU 8%) — downsize from m5.xlarge to t3.large (save $2,400/month). 3 instances are under-provisioned (CPU consistently 95%) — upsize from t3.small to t3.medium to prevent performance issues.*

Q147	What is an AWS Reserved Instance Marketplace?
A	The Reserved Instance Marketplace allows customers to sell unused Standard Reserved Instances to other AWS customers. If business needs change (downsize, close a division), you can recoup remaining RI value. Buyers can purchase RIs with custom terms at potentially lower prices.

▯	*Real-World Example: A company bought 1-year RIs for 100 EC2 instances. After 6 months, they reduce their fleet to 70. They list the remaining 30 RIs on the RI Marketplace (6 months remaining). Another company purchases them at a slight discount — the seller recovers ~$15,000 that would otherwise be wasted.*

Section 17 · AWS Security Deep Dive

Q148	What is AWS Security Hub?
A	AWS Security Hub provides a comprehensive view of security alerts and security posture across AWS accounts. It aggregates findings from GuardDuty, Inspector, Macie, IAM Access Analyzer, and third-party tools. Maps findings to compliance frameworks (CIS, PCI DSS, NIST).
▯	*Real-World Example: A CISO opens Security Hub dashboard: 3 Critical findings (EC2 with public exposure), 12 High findings (unencrypted S3 buckets), 45 Medium findings (unused IAM credentials). All from different AWS services in one unified view — prioritised by severity for the security team.*

Q149	What is IAM Identity Center (SSO)?
A	AWS IAM Identity Center (formerly AWS SSO) provides centralised access management for multiple AWS accounts and cloud applications. Users sign in once and access all assigned accounts and apps. Integrates with Active Directory, Okta, Azure AD. Replaces managing individual IAM users per account.
▯	*Real-World Example: A company has 50 AWS accounts. Without IAM Identity Center: each employee has 50 separate IAM users (50 × password = security nightmare). With IAM Identity Center: one login through company Okta ▯ access to all 50 accounts with assigned permissions. Single sign-on, single password policy.*

Q150	What is AWS Config Rules?
A	AWS Config Rules evaluate whether AWS resource configurations comply with desired settings. AWS provides 300+ managed rules. Examples: s3-bucket-public-read-prohibited, ec2-instance-no-public-ip, rds-instance-public-access-check. Custom rules use Lambda. Non-compliant resources are flagged.

▯	*Real-World Example: A company enables Config Rule: 'encrypted-volumes' — checks all EBS volumes are encrypted. Config continuously evaluates. When a developer creates an unencrypted EBS volume, Config flags it as Non-Compliant within minutes and sends an SNS notification to the security team for remediation.*

Q151	What is Amazon Security Lake?
A	Amazon Security Lake automatically centralises security data from AWS environments, SaaS providers, and on-premises sources into a purpose-built data lake in S3. Data is normalised to the Open Cybersecurity Schema Framework (OCSF). Enables centralised security analytics at scale.

Q152	What is AWS Network Firewall?
A	AWS Network Firewall is a stateful, managed firewall service for VPCs. Provides deep packet inspection, intrusion prevention (IPS), and web filtering. Deployed in a dedicated subnet in each AZ. More advanced than Security Groups (stateful but no deep inspection) and NACLs (stateless).
▯	*Real-World Example: A company needs to block all outbound traffic to known malware C2 servers and all TOR exit nodes from their VPC. AWS Network Firewall with Threat Intelligence managed rules blocks these automatically — a capability not possible with Security Groups or NACLs alone.*

Q153	What is AWS Firewall Manager?
A	AWS Firewall Manager is a security management service for centrally configuring and managing firewall rules across multiple AWS accounts in an AWS Organization. Manages Security Groups, WAF rules, Network Firewall, Shield Advanced, and Route 53 Resolver DNS Firewall from a single account.
▯	*Real-World Example: A company with 100 AWS accounts needs WAF rules applied to all ALBs across all accounts. Without Firewall Manager: manually configure WAF in 100 accounts. With Firewall Manager: create one policy in the management account — automatically applied to all ALBs in all 100 accounts, including new accounts created in the future.*

Q154	What is AWS IAM Access Analyzer?
A	IAM Access Analyzer helps identify resources (S3 buckets, IAM roles, KMS keys, Lambda functions) that are shared with external principals — outside your AWS account or organisation. Continuously monitors and generates findings for unintended external access.

▯	*Real-World Example: IAM Access Analyzer finds: S3 bucket 'employee-data-backup' allows access from AWS account 999999999999 (an external account). The security team investigates — it was a test bucket accidentally made cross-account accessible. They remove the permission immediately. Without Access Analyzer, this could have been undetected for months.*

Q155	What is AWS Resource Access Manager (RAM)?
A	AWS RAM allows sharing of AWS resources with other AWS accounts or within an AWS Organization. Shareable resources include subnets, Transit Gateway, Route 53 Resolver rules, License Manager configurations, and more. Reduces duplication and centralises shared resources.
▯	*Real-World Example: A company creates one shared VPC with shared subnets using RAM. 20 developer accounts can launch EC2 instances into the shared subnets — no need for each team to manage their own VPC/networking. Central networking team maintains the VPC; teams just deploy their applications.*

Section 18 · Serverless & Modern Architectures

Q156	What is AWS SAM?
A	AWS Serverless Application Model (SAM) is an open-source framework for building serverless applications. It extends CloudFormation with simplified syntax for Lambda, API Gateway, DynamoDB, and SQS. SAM CLI enables local testing and debugging of Lambda functions before deploying to AWS.
▯	*Real-World Example: A developer defines their entire serverless app in a SAM template (15 lines of YAML vs 200 lines of CloudFormation). SAM CLI runs the Lambda locally: sam local invoke MyFunction --event test-event.json. After testing, sam deploy pushes to AWS — consistent dev/prod environment.*

Q157	What is AWS Amplify?
A	AWS Amplify is a set of tools and services for building full-stack web and mobile applications. Features: Amplify Hosting (CI/CD for web apps), Amplify Studio (visual app builder), Amplify Libraries (Auth, API, Storage SDKs for React/Vue/iOS/Android), and backend provisioning via CloudFormation.
▯	*Real-World Example: A startup builds a React web app with Amplify: git push ▯ Amplify automatically builds, tests, and deploys to a CDN (like Netlify). The app uses Amplify Auth (Cognito), Amplify API (AppSync/GraphQL), and Amplify Storage (S3) — full-stack app with 50 lines of frontend code, no backend team needed.*

Q158	What is AWS AppRunner?
A	AWS App Runner is a fully managed service for deploying containerised web applications and APIs. Provide source code (GitHub) or a container image — App Runner builds, deploys, scales, and load balances automatically. No VPC configuration, no cluster management.
▯	*Real-World Example: A data scientist builds a Flask ML inference API. They push the Docker image to ECR and point App Runner at it. App Runner deploys the API with HTTPS, auto-scaling (0 to 100 containers), and load balancing in 5 minutes — zero infrastructure knowledge required.*

Q159	What is Lambda Cold Start?
A	A Lambda cold start occurs when a function is invoked for the first time or after being idle — AWS must initialise a new execution environment (load code, start runtime). Cold starts add 100ms–2000ms latency. Mitigation: Provisioned Concurrency (pre-warm containers), use compiled runtimes (Go, Rust), keep packages small.
▯	*Real-World Example: A user reports the first request to a rarely-used Lambda API takes 2 seconds; subsequent requests take 50ms. This is a cold start (2-second init including Java JVM startup). Solution: enable Provisioned Concurrency (5 pre-warmed containers) — all requests now respond in 50ms, including the first.*

Q160	What is Lambda Layers?
A	Lambda Layers allow sharing common code, libraries, and dependencies across multiple Lambda functions. A layer is a ZIP archive stored separately from function code. Functions can reference up to 5 layers. Layers reduce deployment package size and promote code reuse.
▯	*Real-World Example: 10 Lambda functions all use the same 50MB pandas+numpy library. Without layers: each function includes 50MB ZIP = 500MB total deployment storage, slow updates. With a Lambda Layer: one 50MB layer shared by all 10 functions. Update pandas once in the layer — all 10 functions instantly use the new version.*

Q161	What is Lambda@Edge?
A	Lambda@Edge runs Lambda functions in response to CloudFront events (viewer request, viewer response, origin request, origin response) at edge locations globally — closer to users. Used for A/B testing, URL rewriting, HTTP header manipulation, authentication at the edge.

▯	*Real-World Example: A company wants to A/B test a new homepage. Lambda@Edge intercepts every CloudFront viewer request, reads a cookie, and routes 50% of requests to origin A and 50% to origin B — globally, at the edge, with sub-millisecond overhead. No application code changes needed.*

Q162	What is Amazon EventBridge Scheduler?
A	Amazon EventBridge Scheduler is a serverless scheduler for creating, running, and managing scheduled tasks at scale. Supports one-time and recurring schedules. Invokes targets: Lambda, SQS, Step Functions, and 270+ AWS services. More reliable than cron expressions on EC2.
▯	*Real-World Example: A company needs to run a database cleanup Lambda every night at 2AM UTC and a monthly billing report Lambda on the 1st of each month. EventBridge Scheduler handles both with guaranteed delivery — even if Lambda is throttled, the event is retried until successful.*

Q163	What is AWS Glue DataBrew?
A	AWS Glue DataBrew is a visual data preparation tool for cleaning and normalising data without writing code. Data analysts can apply 350+ pre-built transformations (remove duplicates, fix formats, fill missing values) using a point-and-click interface. Profiling shows data quality statistics.
▯	*Real-World Example: A data analyst needs to clean 5 years of messy customer data (inconsistent phone formats, duplicates, missing values). Using DataBrew, they visually apply transformations: standardise phone ▯ (xxx) xxx-xxxx format, remove 3,000 duplicates, fill missing emails with 'unknown'. No Python or SQL coding required.*

Section 19 · Exam Preparation Tips

Q164	What are the CLF-C02 exam domains and weightings?
A	CLF-C02 has 4 domains: 1. Cloud Concepts (24%) — cloud benefits, design principles, migration. 2. Security and Compliance (30%) — shared responsibility, IAM, security services. 3. Cloud Technology and Services (34%) — compute, storage, database, networking, AI/ML. 4. Billing, Pricing, and Support (12%) — pricing models, cost management, support plans.
▯	*Real-World Example: Study allocation: Security (30%) deserves the most attention — master IAM, GuardDuty, Shield, WAF, encryption services. Cloud Technology (34%) covers the most services — broad knowledge over EC2, S3, RDS, VPC. Focus on concepts over CLI commands — CLF-C02 is conceptual, not technical.*

Q165	What is the format of the CLF-C02 exam?
A	CLF-C02 format: 65 questions (50 scored + 15 unscored pilot questions), 90 minutes, passing score 700/1000. Question types: Multiple Choice (1 correct from 4 options) and Multiple Response (2 correct from 5 options). Available in testing centres or online proctored. Cost: $100 USD.
▯	*Real-World Example: Time management tip: 90 minutes for 65 questions = ~83 seconds per question. Flag difficult questions and revisit. Multiple response questions: eliminate clearly wrong answers first. No penalty for guessing — always answer all questions.*

Q166	What is the difference between Availability and Durability?
A	Availability is the percentage of time a service is accessible and operational (e.g., S3 99.99% availability). Durability is the probability that data will not be lost or corrupted (e.g., S3 99.999999999% durability). A service can be unavailable temporarily but your data is still durable — it exists, just not accessible.
▯	*Real-World Example: S3 Standard: 99.99% availability means it could be inaccessible ~52 minutes/year (very rare). 11-nines durability means: store 10 million objects for 10,000 years ▯ expect to lose 1 object. In practice, data is never lost. High availability ≠ high durability — different concepts.*

Q167	What is RPO and RTO in disaster recovery?
A	RPO (Recovery Point Objective) is the maximum acceptable age of data that can be lost in a disaster — how much data loss is acceptable. RTO (Recovery Time Objective) is the maximum acceptable time to restore services after a disaster — how quickly you must recover.
▯	*Real-World Example: A bank's core banking system: RPO = 0 (zero data loss — every transaction must survive), RTO = 5 minutes (must restore within 5 minutes). Implementation: Multi-AZ RDS (synchronous replication, automatic failover in <2 minutes) achieves both. An e-commerce site: RPO = 1 hour (last hourly backup), RTO = 4 hours (acceptable for non-critical).*

Q168	What are the 4 DR strategies in AWS?

A	From lowest cost/slowest recovery to highest cost/fastest: 1. Backup & Restore (RPO hours, RTO hours) — backup to S3/Glacier, restore when needed. 2. Pilot Light (RPO minutes, RTO hours) — minimal core resources always running. 3. Warm Standby (RPO seconds, RTO minutes) — scaled-down full environment. 4. Multi-Site Active/Active (RPO ~0, RTO ~0) — full production in multiple regions.
▯	*Real-World Example: A company chooses Warm Standby: Production in us-east-1 (full scale). In eu-west-1: Auto Scaling group (min 1 instance, normally running), RDS read replica. If us-east-1 fails: promote RDS replica, scale up ASG to full capacity. Recovery: ~15 minutes. Cost: ~30% of running two full environments.*

Q169	What is a Disaster Recovery Pilot Light?
A	Pilot Light is a DR strategy where the critical core of a system is always running in the DR region (like a pilot light on a gas stove). Data is replicated continuously. During a disaster, additional resources (EC2, ELB) are launched around the running core to restore full capacity.
▯	*Real-World Example: Production (us-east-1): Full stack — 20 EC2, 1 ALB, RDS primary. DR (eu-west-1) Pilot Light: Only RDS replica running (minimal cost $150/month). Disaster: us-east-1 fails. Response: Promote RDS replica (5 min), launch 20 EC2 + ALB (10 min). Total RTO: ~15 minutes. Cost vs full standby: 10x cheaper.*

Q170	What is the CAF Migration Readiness Assessment?
A	AWS Migration Readiness Assessment (MRA) is a structured review of an organisation's readiness to migrate to AWS. Evaluates capabilities across all 6 CAF perspectives. Results in a readiness score and action plan identifying gaps to address before migration.

Q171	How do you calculate EC2 costs?
A	EC2 cost = Instance hours × hourly rate. Factors: instance type (t3.micro vs r5.4xlarge), pricing model (on-demand, reserved, spot), region (us-east-1 is cheapest), OS (Windows > Linux), and additional costs (EBS storage, data transfer, Elastic IPs). Use the AWS Pricing Calculator for estimates.
▯	*Real-World Example: m5.xlarge Linux, us-east-1, on-demand: $0.192/hr × 720 hrs = $138.24/month. Same instance, 1-year All-Upfront Reserved: $1,022/year = $85.17/month — 38% savings. Same instance, Spot (current price ~$0.06/hr): $0.06 × 720 = $43.20/month — 69% savings (if interruptions are acceptable).*

Q172	What should you know about AWS pricing for S3?
A	S3 pricing has several components: Storage ($0.023/GB/month for Standard), Requests (GET $0.0004 per 1,000, PUT $0.005 per 1,000), Data Transfer Out (first 100GB free, then $0.09/GB), Storage Class Analysis, Replication. Inbound data transfer is always free. Prices vary slightly by region.
□	*Real-World Example: A company stores 10TB in S3 Standard: $0.023 × 10,240 GB = $235/month. They serve 1TB/day to users via CloudFront: 1,000 GB × $0.009 (CloudFront price cheaper than S3 direct at $0.085) = $9/day. Key insight: CloudFront + S3 is cheaper than S3 direct for high-traffic content delivery.*

Q173	What is the AWS Well-Architected Tool?
A	The AWS Well-Architected Tool is a free service in the AWS Console. You define a workload, answer 50-80 questions across the 6 pillars, and receive a report of High-Risk Issues (HRIs) and Medium-Risk Issues (MRIs) with detailed improvement guidance. Helps teams build better architectures proactively.
□	*Real-World Example: A team answers: 'Do you use Multi-AZ for your databases?' □ No □ HRI flagged. 'Do you have automated backups?' □ No □ HRI flagged. After addressing all HRIs, they re-run the review and achieve a 'Well-Architected' status — required by many enterprise customers as a vendor prerequisite.*

Q174	What key differences distinguish the AWS Enterprise Support plan?
A	Enterprise Support ($15,000+/month) provides: dedicated Technical Account Manager, Concierge Support Team for billing, <15 minute response for business-critical system down, Infrastructure Event Management (for launches and migrations), Well-Architected Reviews, and access to all Trusted Advisor checks plus custom checks.
□	*Real-World Example: A major bank plans to migrate 500 applications to AWS. They use Enterprise Support's Infrastructure Event Management — a dedicated AWS team works with them during the 6-month migration, providing 24/7 support, architecture guidance, and monitoring. Worth the $15K/month on a $50M migration.*

Q175	What is AWS OpsWorks?
A	AWS OpsWorks is a configuration management service providing managed instances of Chef and Puppet — automation platforms for server configuration management. OpsWorks Stacks lets you model applications as stacks and layers. Largely replaced by Systems Manager for new workloads.

Q176	What is the difference between AWS Config and AWS CloudTrail?
A	CloudTrail answers 'Who did what when?' — it records API calls made to AWS services (action log). Config answers 'What does my infrastructure look like over time?' — it records the state of resource configurations and when they changed (state log). CloudTrail logs actions; Config logs configurations.
▯	*Real-World Example: CloudTrail: IAM user 'alice' called ModifyDBInstance on RDS 'prod-db' at 14:30 to change instance class from db.t3.medium to db.t3.large. Config: RDS 'prod-db' configuration history shows instance class changed from db.t3.medium to db.t3.large at 14:32. Both logs are complementary for full auditability.*

Section 20 · Advanced Topics & Miscellaneous

Q177	What is Serverless computing?
A	Serverless computing means you run code without provisioning or managing servers. AWS manages the infrastructure, scaling, and availability automatically. You pay only for actual execution. AWS serverless services: Lambda, Fargate, API Gateway, DynamoDB, S3, SNS, SQS, Step Functions.
▯	*Real-World Example: Traditional: Developer manages EC2 server, installs Node.js, configures nginx, monitors CPU, patches OS — even when no requests come in. Serverless: Developer writes a Lambda function. AWS handles everything else. When there are 0 requests, cost = $0.*

Q178	What is Infrastructure as Code (IaC)?
A	IaC is the practice of managing infrastructure through code files rather than manual processes. Benefits: version control, consistency, repeatability, auditability, disaster recovery (redeploy from code). AWS IaC tools: CloudFormation (AWS native), AWS CDK (programmatic), Terraform (third-party).
▯	*Real-World Example: A company's production environment was accidentally deleted. Without IaC, rebuilding takes weeks. With CloudFormation templates in Git, the entire environment (VPC, subnets, EC2, RDS, ALB) is rebuilt in 25 minutes using one command: aws cloudformation deploy.*

Q179	What is Containerisation?

A	Containerisation packages an application with its dependencies, libraries, and configuration into a portable container image (Docker). Containers run consistently across any environment (developer laptop, staging, production). AWS container services: ECS, EKS, Fargate, ECR.
□	*Real-World Example: A Node.js app works on the developer's Mac but fails on the Linux production server ('works on my machine' problem). Containerising the app with Docker eliminates this — the exact same container image runs identically on the developer's laptop, CI/CD, and production ECS.*

Q180	What is Amazon ECR?
A	Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry for storing, managing, and deploying container images. It integrates with ECS, EKS, and CodeBuild. Images are encrypted, and IAM controls access. Supports vulnerability scanning.
□	*Real-World Example: A CI/CD pipeline: CodeBuild builds a Docker image □ pushes to ECR □ ECS pulls the image from ECR and deploys new containers. ECR stores every version tagged by Git commit hash — teams can roll back to any previous version in seconds.*

Q181	What is AWS Outposts?
A	AWS Outposts is a fully managed service that extends AWS infrastructure, services, and tools to any data center or on-premises facility. AWS installs physical hardware at your location. You use AWS APIs and console to manage on-prem resources just like cloud resources.
□	*Real-World Example: A hospital needs to process MRI scan data locally (data sovereignty law prohibits sending patient data to the cloud). They install AWS Outposts in their data center. ML models run locally on Outposts hardware using the same SageMaker APIs used in the cloud.*

Q182	What is AWS Local Zones?
A	AWS Local Zones extend AWS infrastructure to geographic locations closer to large population centres. They enable single-digit millisecond latency for end users. Use cases: media & entertainment, real-time gaming, live video streaming, ML inference.

▯	*Real-World Example: A real-time online gaming company needs <5ms latency for Los Angeles players. AWS Los Angeles Local Zone puts compute within 5ms of LA users. The same game runs on AWS us-west-2 region for other users — Local Zone serves only the latency-sensitive LA audience.*

Q183	What is AWS Wavelength?
A	AWS Wavelength embeds AWS compute and storage within telecom providers' 5G networks. Applications can reach speeds of <10ms latency for mobile devices. Use cases: autonomous vehicles, AR/VR, connected factory floor, smart cities.
▯	*Real-World Example: An AR app for a stadium: fans point their phone at a player and see live stats overlaid. With AWS Wavelength in the 5G network, processing happens within 5ms (vs 80ms over the regular internet) — the overlay appears instantaneous.*

Q184	What is AWS Greengrass?
A	AWS IoT Greengrass extends AWS cloud capabilities to edge devices (local servers, smart devices, IoT gateways). Devices can run Lambda functions, ML inference, and data sync locally even without internet connectivity — syncing to AWS when connected.
▯	*Real-World Example: Wind turbines in a remote field use Greengrass to run ML models locally that predict blade failures in real-time. Even when the satellite internet link is down, the model continues running. Data syncs to AWS S3 when connectivity resumes.*

Q185	What is Amazon Connect?
A	Amazon Connect is a cloud-based contact centre service. Set up a call centre in minutes — no hardware, no contracts. Features: natural language IVR (Lex integration), real-time and historical analytics, CRM integration (Salesforce), agent desktop, call recording (S3).
▯	*Real-World Example: A startup launches a customer support line using Amazon Connect in 2 hours. They configure an IVR in the visual designer, route calls to remote agents' laptops via browser, and record all calls to S3 for quality assurance — no PBX hardware purchased.*

Q186	What is Amazon Chime?
A	Amazon Chime is a communications service for online meetings, video conferencing, calls, and chat. Chime SDK allows developers to embed real-time audio/video into their own applications. Pay-per-use (no upfront costs).

Q187	What is AWS AppSync?
A	AWS AppSync is a managed GraphQL service. It connects to DynamoDB, Lambda, RDS, HTTP APIs, and Elasticsearch as data sources. Supports real-time data with subscriptions (WebSockets). Ideal for building mobile and web apps requiring real-time sync.
▯	*Real-World Example: A collaborative document editing app uses AppSync subscriptions. When User A edits a paragraph, AppSync broadcasts the change to all connected users in real-time via WebSocket — similar to Google Docs collaboration, built with minimal backend code.*

Q188	What is Amazon Elastic Transcoder?
A	Amazon Elastic Transcoder converts media files stored in S3 to formats suitable for different devices (phones, tablets, desktops). You create pipelines and jobs specifying the source and output format. Billed per minute of output.
▯	*Real-World Example: A video platform receives raw 4K .mov files from creators. Elastic Transcoder converts each file to: 4K MP4 (desktop), 1080p MP4 (HD), 720p MP4 (mobile), 480p MP4 (low bandwidth) — 4 output files ready for CloudFront CDN distribution.*

Q189	What is the difference between AWS and traditional IT?
A	Traditional IT: High CapEx (buy servers), long procurement cycles (weeks/months), fixed capacity (over-provision or under-provision), you manage physical security and hardware. AWS: OpEx (pay per use), instant provisioning (seconds), elastic capacity, AWS manages physical infrastructure.
▯	*Real-World Example: Traditional: Company buys 10 servers for expected peak load, using 30% capacity normally (wasted). AWS: Run 3 EC2 instances normally, Auto Scale to 10 during peak, scale back to 3 after. Pay only for actual usage — potentially 70% cost savings.*

Q190	What is Amazon WorkSpaces?
A	Amazon WorkSpaces is a fully managed Desktop-as-a-Service (DaaS) solution. Provides cloud-based Windows or Linux desktops accessible from any device and location. Eliminates the need to manage physical desktop hardware. Pay monthly or hourly.

▯	*Real-World Example: A financial company's 500 employees work from home during the pandemic. They provision 500 WorkSpaces in 2 hours — each employee gets a full Windows desktop with all business apps, accessible from their home computer via browser. No VPN complexity.*

Q191	What is Amazon AppStream?
A	Amazon AppStream 2.0 is a fully managed application streaming service. Stream individual desktop applications (not full desktops) to any browser. No installation required. Used for software trials, remote access to engineering tools, educational software.
▯	*Real-World Example: A school wants students to use AutoCAD without buying software licences or powerful computers. AppStream streams AutoCAD from AWS to students' Chromebooks via a browser. Students get full AutoCAD performance without any local installation.*

Q192	What is AWS DataSync?
A	AWS DataSync is an automated data transfer service that moves data between on-premises storage (NFS, SMB, HDFS) and AWS storage services (S3, EFS, FSx). It accelerates transfers using network optimisation, compression, and encryption. Monitors and validates transfers.
▯	*Real-World Example: A media company has 200TB of video assets on NAS (NFS). DataSync transfers to S3 at 10x the speed of a simple copy script — using parallelism, checksums to verify integrity, and scheduling to run during off-peak hours without manual oversight.*

Q193	What is Amazon FSx?
A	Amazon FSx provides fully managed file systems for specific workloads: FSx for Windows File Server (SMB, NTFS — for Windows workloads), FSx for Lustre (high-performance computing, ML training), FSx for NetApp ONTAP, FSx for OpenZFS.
▯	*Real-World Example: A media company running DaVinci Resolve video editing on Windows needs shared fast storage. FSx for Windows File Server provides SMB shares accessible from 50 Windows EC2 instances simultaneously, with Active Directory integration for access control.*

Q194	What is Amazon Macie?

A	Amazon Macie is a data security service that uses machine learning to automatically discover, classify, and protect sensitive data in S3. It identifies PII (names, credit card numbers, SSNs), proprietary data, and access anomalies.
▯	*Real-World Example: A company has 10,000 S3 buckets. Macie scans them and finds: Bucket 'old-backup-2019' contains 50,000 files with credit card numbers — publicly accessible. The security team is alerted and immediately makes the bucket private and encrypts the files.*

Q195	What is Amazon Detective?
A	Amazon Detective analyses, investigates, and identifies the root cause of security findings. It collects and processes data from VPC Flow Logs, CloudTrail, and GuardDuty findings automatically, creating visualisations of resource behaviour over time.
▯	*Real-World Example: GuardDuty flags that EC2 instance i-12345 is doing port scanning. Amazon Detective shows: this instance started unusual behaviour 3 days ago, immediately after an IAM user from IP 203.0.113.10 logged in — tracing the intrusion timeline clearly.*

Q196	What is the AWS Acceptable Use Policy?
A	AWS Acceptable Use Policy (AUP) prohibits use of AWS services for illegal, harmful, or abusive activities, including: spam, port scanning, DDoS attacks, distributing malware, child exploitation, copyright infringement, and activities that violate others' rights. All customers must agree to the AUP.

Q197	What is the AWS Service Level Agreement (SLA)?
A	AWS SLAs define the uptime commitments for services. Example: EC2 SLA = 99.99% monthly uptime. If AWS fails to meet the SLA, customers receive service credits. EC2 99.99% = max 52.6 minutes downtime/year. SLAs vary by service and are detailed at aws.amazon.com/legal/service-level-agreements.
▯	*Real-World Example: AWS fails to meet the EC2 99.99% SLA — actual uptime was 99.8% (about 87 minutes downtime). The customer is eligible for a 10% service credit on their monthly EC2 bill for that region. Credits are applied to future bills, not cash refunds.*

Q198	What is Amazon Forecast?

A	Amazon Forecast is a fully managed ML service for time-series forecasting. It uses the same technology Amazon uses for demand forecasting in its own supply chain. No ML expertise needed — provide historical data and it generates forecasts.
▯	*Real-World Example: A retailer provides 3 years of weekly sales data to Amazon Forecast. The service trains multiple ML models, selects the best, and generates item-level demand forecasts for the next 90 days. The procurement team uses this to optimise inventory — reducing overstock by 25%.*

Q199	What is Amazon Personalize?
A	Amazon Personalize is a fully managed ML service for building real-time personalisation and recommendation systems. No ML expertise needed. Provide user activity data (clicks, views, purchases) and item metadata, and Personalize trains and hosts a recommendation model.
▯	*Real-World Example: A streaming service uses Amazon Personalize trained on viewing history. When a user opens the app, Personalize returns their personal top-10 recommended titles in <100ms. Click-through rate increases 35% compared to rule-based recommendations.*

Q200	What is AWS DeepRacer?
A	AWS DeepRacer is a 1/18th scale autonomous racing car and online racing simulator for learning reinforcement learning (RL). Developers train RL models in a simulated 3D track, then deploy to the physical car. AWS DeepRacer League is a global competitive racing series.

Q201	What is AWS Artifact?
A	AWS Artifact is a self-service portal providing on-demand access to AWS compliance reports and security certifications (SOC 1/2/3, PCI DSS, ISO 27001, HIPAA). Also for reviewing, accepting, and managing AWS agreements. Free to use. Essential for compliance audits.
▯	*Real-World Example: A startup is going through a SOC 2 audit. Their auditor asks for AWS's SOC 2 report. The team downloads it from AWS Artifact in 2 minutes — no need to contact AWS support or wait for documents. The auditor accepts it as evidence of AWS's security controls.*

Q202	What is AWS Compliance?

A AWS maintains compliance with global security standards: PCI DSS (payments), HIPAA (US healthcare), GDPR (EU data privacy), SOC 1/2/3 (financial controls), ISO 27001 (information security), FedRAMP (US federal government). Customers can build compliant workloads on AWS.

□ *Real-World Example: A US hospital building an electronic health record app on AWS chooses HIPAA-eligible services (RDS, S3, Lambda). AWS signs a BAA (Business Associate Agreement). The hospital is responsible for configuring those services correctly — shared responsibility for HIPAA compliance.*

Q203 What is the difference between Horizontal and Vertical Scaling?

A Vertical Scaling (Scale Up/Down): increase/decrease the resources of an existing instance (e.g., EC2 t3.medium □ t3.xlarge). Has a ceiling (largest instance size). Requires downtime. Horizontal Scaling (Scale Out/In): add/remove instances. No ceiling. No downtime. Preferred for cloud-native apps.

□ *Real-World Example: Database server struggling: Vertical scaling — stop RDS instance, change to larger instance type (5 minutes downtime). App server struggling: Horizontal scaling — Auto Scaling adds 5 more EC2 instances in 2 minutes — zero downtime, users don't notice.*

Q204 What is Multi-AZ Deployment in RDS?

A RDS Multi-AZ creates a synchronous standby replica in a different AZ. If the primary fails, RDS automatically fails over to the standby in 1-2 minutes — no manual intervention. Standby is not used for reads (unlike Read Replicas) — it's only for failover.

□ *Real-World Example: An RDS MySQL primary instance in us-east-1a fails (hardware failure). RDS detects failure and promotes the standby in us-east-1b within 90 seconds. The application's connection string (DNS) automatically points to the new primary. The on-call engineer wakes up to find the issue already resolved.*

Q205 What is RDS Read Replica?

A An RDS Read Replica is an asynchronous copy of the primary database used to offload read traffic. Unlike Multi-AZ (failover only), Read Replicas serve actual read queries. You can have up to 15 replicas (Aurora). Can be in different regions (cross-region Read Replicas).

▯	*Real-World Example: An analytics team runs heavy SQL reports on the production database, causing 80% CPU utilisation. Adding a Read Replica allows reports to run against the replica. Production database CPU drops to 20% — no impact on transactional users.*

Q206	What is Amazon S3 Transfer Acceleration?
A	S3 Transfer Acceleration uses CloudFront edge locations to speed up uploads to S3. When you upload to your S3 bucket's accelerated endpoint (bucketname.s3-accelerate.amazonaws.com), the file is routed to the nearest edge location, then transferred over AWS's fast backbone to S3.
▯	*Real-World Example: A photographer in Sydney uploads 10GB RAW files to an S3 bucket in us-east-1. Normal upload: 45 minutes. With Transfer Acceleration: 18 minutes — the file enters AWS's backbone at the Sydney edge location, traversing AWS's high-speed private network to Virginia.*

Q207	What is Cross-Region Replication (CRR) in S3?
A	S3 Cross-Region Replication automatically replicates objects from a source bucket in one region to a destination bucket in another region. Requires versioning enabled in both buckets. Used for compliance (data residency), disaster recovery, and reducing latency.
▯	*Real-World Example: A multinational company stores all data in us-east-1 (primary). S3 CRR replicates everything to eu-west-1. If a catastrophic failure destroys the US region's data (extremely unlikely), the EU copy remains intact. This is a multi-region disaster recovery strategy.*

Q208	What is AWS Elastic IP?
A	An Elastic IP (EIP) is a static public IPv4 address allocated to your AWS account. Unlike standard public IPs (which change when EC2 restarts), EIPs persist. You can remap EIPs to different EC2 instances. AWS charges for EIPs that are NOT associated with a running instance.
▯	*Real-World Example: A company runs a server at 203.0.113.10 (EIP). The server has a hardware issue. They stop the instance, launch a new EC2, reassociate the EIP to the new instance. External users continue accessing 203.0.113.10 — seamless cutover with no DNS change needed.*

Q209	What is AWS Global Accelerator?

A	AWS Global Accelerator is a networking service that uses the AWS global network to route traffic from users to the nearest AWS edge location and then over the AWS backbone to your application (EC2, ELB, EIP). Provides static anycast IPs and automatic failover. Different from CloudFront (no caching).
▯	*Real-World Example: A gaming leaderboard API is deployed in us-east-1 and eu-west-1. Global Accelerator routes a Singapore player to the us-east-1 endpoint over the AWS backbone (faster than internet). If us-east-1 fails, it automatically fails over to eu-west-1 in under 30 seconds.*

Q210	What is Amazon Elastic IP vs Public IP?
A	A Public IP is automatically assigned when EC2 starts (if enabled) but is released when stopped — it changes every restart. An Elastic IP is a static public IP you allocate to your account and manually associate. EIP persists across stops/starts. Use EIP when external systems (DNS, firewall whitelists) require a consistent IP.

Q211	What is VPC Flow Logs?
A	VPC Flow Logs capture information about IP traffic going to and from network interfaces in your VPC. Stored in CloudWatch Logs or S3. Used for security analysis, troubleshooting connectivity, and compliance. Can be enabled at VPC, subnet, or network interface level.
▯	*Real-World Example: An app can't connect to the RDS database. VPC Flow Logs show: EC2 sends TCP SYN to RDS port 3306 ▯ REJECT. This reveals the RDS security group doesn't allow port 3306 from the EC2's security group — root cause identified in 2 minutes.*

Q212	What is an AWS Launch Template?
A	A Launch Template specifies EC2 configuration parameters (AMI, instance type, key pair, security groups, user data, EBS volumes) saved as a versioned template. Used by Auto Scaling groups and EC2 Fleet. Replaces Launch Configurations with added versioning and parameter override capabilities.
▯	*Real-World Example: A DevOps team creates a Launch Template for their web servers: Amazon Linux 2, t3.medium, specific security group, 50GB gp3 EBS, User Data script. The Auto Scaling group uses this template — all new instances launched are identical. Template v2 updates to t3.large without disrupting running instances.*

Q213	What is Amazon Lightsail vs EC2?

A Lightsail is a simplified, fixed-price virtual server for straightforward use cases (WordPress, dev environments, simple web apps). EC2 is the full-featured compute service with hundreds of instance types, fine-grained networking, and integration with all AWS services. Lightsail is easier but less flexible.

□ *Real-World Example: Blog with predictable traffic □ Lightsail $5/month (simple, no AWS expertise needed). Enterprise e-commerce with Auto Scaling, custom VPC, direct RDS integration □ EC2 (full AWS control required).*

Q214 What is AWS Trusted Advisor and what categories does it cover?

A AWS Trusted Advisor provides automated best practice recommendations across 5 categories: 1. Cost Optimization (unused resources, right-sizing). 2. Performance (EC2 utilisation, CloudFront configuration). 3. Security (open ports, IAM usage, MFA). 4. Fault Tolerance (AZ redundancy, RDS backups). 5. Service Limits (approaching AWS limits).

□ *Real-World Example: Trusted Advisor Security check finds: 3 IAM users haven't rotated their access keys in 180+ days and the root account has no MFA. The security team remediates both in 30 minutes — preventing potential credential-based attacks.*

Q215 What is Amazon CloudWatch Logs Insights?

A CloudWatch Logs Insights is an interactive log analytics service. Use a SQL-like query language to search and analyse logs from Lambda, EC2, ECS, API Gateway etc. Queries run in seconds across billions of log events. Visualise results as time series graphs or tables.

□ *Real-World Example: An API is returning 500 errors. The developer queries Lambda logs: fields @timestamp, errorMessage | filter @message like 'ERROR' | stats count() by bin(5m). They see errors spiked at 14:30 — correlating with a deployment at 14:28. Root cause found in 2 minutes.*

Q216 What is the difference between CloudWatch and CloudTrail?

A CloudWatch monitors performance metrics and logs (CPU utilisation, Lambda duration, application logs) — answers 'How is my infrastructure performing?'. CloudTrail records API activity and user actions (who did what, when) — answers 'Who changed what in my AWS account?'. Both are complementary security and operations tools.

▯	*Real-World Example: Database CPU spiked to 100% at 2PM ▯ investigate with CloudWatch metrics. Someone deleted a production S3 bucket at 2PM ▯ investigate with CloudTrail logs. Often used together: CloudWatch alarm triggers investigation, CloudTrail identifies the human cause.*

Q217	What is Amazon EventBridge vs SNS vs SQS?
A	SQS is a message queue for decoupling services (producer ▯ queue ▯ consumer). SNS is a pub-sub notification service (topic ▯ multiple subscribers simultaneously). EventBridge is an event bus for event-driven architectures with advanced filtering, scheduling, and 90+ AWS service integrations. Often used together.
▯	*Real-World Example: Architecture: EC2 metrics ▯ EventBridge rule (CPU > 90%) ▯ SNS topic ▯ email to ops team AND SQS queue ▯ Lambda reads queue and scales up EC2. All three used together: EventBridge (event routing), SNS (fan-out notification), SQS (reliable async processing).*

Q218	What is Blue/Green Deployment?
A	Blue/Green deployment is a release strategy where two identical production environments (Blue = current, Green = new version) run simultaneously. Traffic is gradually shifted from Blue to Green. If the new version has issues, traffic is instantly routed back to Blue (rollback). Eliminates downtime.
▯	*Real-World Example: A company deploys a new API version (Green) alongside the current version (Blue). They use Route 53 weighted routing: 10% ▯ Green, 90% ▯ Blue. After 30 minutes with no errors, they shift to 100% ▯ Green. Total deployment time: 40 minutes, zero downtime, instant rollback available throughout.*

Q219	What is a Canary Deployment?
A	A canary deployment gradually rolls out a new software version to a small subset of users before deploying to everyone. Named after the canary-in-a-coal-mine concept. Used in AWS CodeDeploy, Lambda, API Gateway. If errors spike in the canary group, the deployment is automatically rolled back.
▯	*Real-World Example: A company deploys a Lambda function update to 5% of traffic (canary). CloudWatch monitors error rates for 10 minutes. Error rate is 0.1% (baseline) — automatic approval proceeds deployment to 100%. If error rate > 1%, automatic rollback to previous version with zero customer impact.*

Q220 What is AWS Backup?

A AWS Backup is a centralized, fully managed backup service for AWS resources: EBS volumes, RDS databases, DynamoDB tables, EFS, FSx, EC2 instances, S3. Create backup plans with schedules and retention policies. Store backups in AWS Backup Vaults (cross-account and cross-region).

□ *Real-World Example: A company creates a backup plan: daily backups of all RDS instances (7-day retention), weekly backups of EFS (30-day retention), monthly backups of all EBS volumes (1-year retention). AWS Backup executes this automatically — meeting compliance requirements with zero manual work.*

Q221 What is Amazon Elastic Cache vs DAX?

A ElastiCache (Redis/Memcached) is a general-purpose in-memory cache for any application — reduce database load, cache sessions, leaderboards. DAX (DynamoDB Accelerator) is a purpose-built in-memory cache for DynamoDB only — reduces read latency from milliseconds to microseconds. Both use in-memory caching but for different use cases.

□ *Real-World Example: Web app using MySQL □ add ElastiCache Redis to cache query results. DynamoDB app needing microsecond response □ add DAX in front of DynamoDB. They cannot be swapped — DAX works only with DynamoDB APIs.*

Q222 What is AWS Certificate Manager (ACM)?

A AWS Certificate Manager provisions, manages, and deploys SSL/TLS certificates. Free for public certificates used with AWS services (ALB, CloudFront, API Gateway). Handles automatic renewal — no more certificate expiry surprises. Also supports private CAs for internal certificates.

□ *Real-World Example: A company's e-commerce site needs HTTPS. They request a free certificate from ACM for checkout.company.com. ACM validates domain ownership via DNS (Route 53), issues the certificate, and attaches it to the ALB. Auto-renews every 13 months — the team never thinks about certificates again.*

Q223 What is Amazon VPC Endpoints?

A VPC Endpoints allow private connectivity from your VPC to AWS services (S3, DynamoDB, EC2 API, etc.) without an internet gateway, NAT gateway, or VPN. Two types: Interface Endpoints (uses PrivateLink, ENI in your VPC) and Gateway Endpoints (for S3 and DynamoDB, free).

▯	*Real-World Example: EC2 instances in a private subnet need to access S3. Without a VPC Endpoint, traffic routes through a NAT Gateway (cost: $0.045/GB). With a Gateway Endpoint, S3 traffic stays within AWS's private network for free. For a company transferring 100TB/month: saves $4,500/month.*

Q224	What is AWS PrivateLink?
A	AWS PrivateLink provides private connectivity between VPCs and AWS services or third-party services hosted in AWS. Traffic never traverses the public internet. Service providers expose services through Interface Endpoints; consumers access them through their VPC.
▯	*Real-World Example: Company A offers an API service. Company B wants to call it privately. Company A creates a PrivateLink service. Company B creates an Interface VPC Endpoint in their VPC. Company B's EC2 instances call Company A's API over a private IP — no traffic leaves AWS's network.*

Q225	What is Amazon SWF?
A	Amazon Simple Workflow Service (SWF) is a workflow coordination service for building distributed applications with tasks that execute sequentially or in parallel. Now largely replaced by AWS Step Functions. SWF has a maximum workflow duration of 1 year.

Q226	What is Amazon Mechanical Turk?
A	Amazon Mechanical Turk (MTurk) is a crowdsourcing marketplace for human intelligence tasks — tasks that require human judgement. Used for data labeling (ML training), survey data collection, content moderation review, and transcription. Businesses post 'HITs' (Human Intelligence Tasks).
▯	*Real-World Example: An ML team needs 100,000 images labeled with bounding boxes (for object detection training). They post the task on MTurk. 10,000 workers worldwide label images over 48 hours for $0.10/image = $10,000 — far cheaper and faster than internal teams.*

Q227	What is Amazon QuickSight Q?
A	Amazon QuickSight Q is a machine learning-powered natural language feature in QuickSight. Business users type questions in plain English ('What were total sales last quarter by region?') and QuickSight Q generates the visualisation automatically — no SQL or BI skills needed.

Q228	What are AWS Service Limits (Quotas)?
A	AWS imposes default limits on resources to prevent accidental overspending and protect infrastructure. Examples: 32 vCPUs per region (EC2 on-demand), 100 S3 buckets per account, 5 VPCs per region. Limits can be increased by submitting a Service Quota increase request via the AWS console.
▯	*Real-World Example: A company's Auto Scaling group needs 100 EC2 instances during a product launch but hits the default limit of 32 vCPUs. They submit a Service Quota increase request 2 weeks before launch. AWS approves: new limit 200 vCPUs. The launch succeeds without capacity issues.*

Q229	What is Amazon Connect vs Chime?
A	Amazon Connect is a cloud contact centre for customer-facing call centers (IVR, agent routing, call recording). Amazon Chime is an internal communications tool (video meetings, chat, calls) for employee collaboration. They serve different audiences: Connect = customer service, Chime = employee communication.

Q230	What is AWS Migration Hub?
A	AWS Migration Hub provides a central location to track the progress of application migrations across multiple AWS migration tools (DMS, Application Migration Service, Server Migration Service). Provides a consolidated view of migration status across all migrating applications.
▯	*Real-World Example: A company migrating 200 applications uses DMS for databases and Application Migration Service for servers. Migration Hub shows a single dashboard: 45 apps completed, 130 in progress, 25 not started — with detailed status for each, all in one view.*

Q231	What is Amazon Translate vs Polly vs Transcribe?
A	Amazon Translate converts text between languages (English ▯ Hindi). Amazon Polly converts text to speech (text ▯ MP3 audio). Amazon Transcribe converts speech to text (audio ▯ text). They can be chained: Transcribe (speech to text) ▯ Translate (English to Spanish) ▯ Polly (Spanish speech output).
▯	*Real-World Example: A customer service recording in English: Transcribe converts audio to English text ▯ Translate converts to Hindi text ▯ Polly reads the Hindi text aloud. This real-time pipeline enables an English-speaking agent to 'speak' in Hindi to an Indian customer.*

Q232	What is Amazon Kendra?
A	Amazon Kendra is an intelligent enterprise search service powered by ML. It understands natural language queries and returns precise answers from your internal documents (SharePoint, S3, RDS, ServiceNow). Unlike keyword search, Kendra understands intent.
▯	*Real-World Example: An employee asks the company intranet: 'What is our parental leave policy?' Traditional search returns 50 documents mentioning 'parental' and 'leave'. Kendra reads the HR policy document and returns: 'Employees are entitled to 16 weeks of paid parental leave' — the exact answer from page 23.*

Q233	What is Amazon Fraud Detector?
A	Amazon Fraud Detector is a fully managed ML service for identifying potentially fraudulent online activities. Pre-built fraud detection models trained on Amazon's own 20+ years of fraud detection experience. Use cases: online payment fraud, fake account creation, loyalty program abuse.
▯	*Real-World Example: An online marketplace uses Fraud Detector for account registration. When a new user signs up, Fraud Detector analyses: email domain, IP address velocity (same IP creating 50 accounts/hour), device fingerprint, and similar patterns. Fraudulent accounts are blocked before any harm occurs.*

Section 21 · Networking Advanced

Q234	What is AWS Transit Gateway?
A	AWS Transit Gateway is a network transit hub that connects multiple VPCs and on-premises networks through a central hub. Instead of creating a mesh of VPC peering connections, Transit Gateway acts as a router. Supports up to 5,000 VPCs, multicast, and inter-region peering.
▯	*Real-World Example: A company has 20 VPCs (dev, staging, prod, shared-services, security, etc.). Without TGW: 190 peering connections needed. With Transit Gateway: 20 attachments to one TGW hub. Adding a new VPC requires 1 connection, not 20.*

Q235	What is AWS Route 53 Failover routing?
A	Route 53 Failover routing directs traffic to a primary resource when it is healthy, and to a secondary resource (standby) when the primary is unhealthy. Health checks monitor the primary endpoint. Failover is automatic — DNS switches within 30-60 seconds of a failed health check.

▯	*Real-World Example: Primary EC2 web server (us-east-1) + S3 static error page (secondary). Route 53 health check pings port 80 every 30 seconds. Server goes down ▯ 2 consecutive failed checks ▯ Route 53 starts returning the S3 error page IP. Users see a maintenance page instead of browser errors.*

Q236	What is an AWS NAT Instance vs NAT Gateway?
A	NAT Instance: An EC2 instance configured to do NAT (older approach). You manage it — patching, scaling, HA. NAT Gateway: A managed AWS service — highly available within an AZ, automatically scales, no patching. NAT Gateway is preferred for new architectures; NAT Instances are legacy but cheaper for very low traffic.

Q237	What is AWS PrivateLink vs VPC Peering?
A	VPC Peering creates a full network connection between two VPCs — all resources can reach each other (bidirectional). PrivateLink exposes a specific service from one VPC to consumers — consumers can only access that specific endpoint, not the entire VPC. PrivateLink is more secure for SaaS-style service exposure.
▯	*Real-World Example: Company A sells a data API. VPC Peering with 100 customers would let customers scan Company A's entire VPC. PrivateLink: customers only see the API endpoint IP — no access to Company A's internal resources. PrivateLink is the correct architectural choice for this use case.*

Q238	What is AWS Global Accelerator vs CloudFront?
A	CloudFront caches content at edge locations (CDN) — reduces latency by serving cached content near users. Global Accelerator uses the AWS global network to route traffic to the nearest edge, then over AWS backbone to your origin — reduces latency for dynamic/uncacheable content. CloudFront for static content; Global Accelerator for dynamic APIs.

Q239	What is a Bastion Host?
A	A Bastion Host (jump server) is an EC2 instance in a public subnet that provides secure access to instances in private subnets. You SSH into the bastion, then SSH from the bastion to private instances. AWS Systems Manager Session Manager is the modern replacement — no bastion needed.

□	*Real-World Example: Traditional: SSH to bastion (54.123.45.67) □ SSH from bastion to private DB server (10.0.2.100). Modern: Use Systems Manager Session Manager — connect to private instances directly through the AWS console or CLI, no bastion EC2 cost, no SSH ports open, full audit trail.*

Section 22 · Data Services

Q240	What is Amazon OpenSearch Service?
A	Amazon OpenSearch Service is a managed search and analytics engine (formerly Elasticsearch Service). Used for log analytics, full-text search, and monitoring. Integrates with Kinesis Firehose, Lambda, CloudWatch Logs. Provides OpenSearch Dashboards (formerly Kibana) for visualisation.
□	*Real-World Example: An e-commerce company ingests all CloudFront logs via Kinesis Firehose into OpenSearch. A dashboard shows real-time search terms, error rates, and geographic traffic. When error rates spike, engineers query: filter status=503 AND timestamp>NOW-1h — results in seconds across billions of log events.*

Q241	What is Amazon EMR?
A	Amazon Elastic MapReduce (EMR) is a cloud big data platform for processing vast data using Apache Spark, Hadoop, Hive, and Flink. Runs on EC2 clusters. EMR manages provisioning, scaling, and Hadoop configuration. Uses S3 as the primary data store.
□	*Real-World Example: A company processes 10TB of daily clickstream logs. EMR launches a 20-node Spark cluster in 5 minutes, runs the job (1 hour), writes results to S3, terminates. Cost: 20 nodes × $0.10/hr × 1 hr = $2. Running on a persistent server = $1,440/month. EMR saves 99.9% for batch workloads.*

Q242	What is Amazon MSK (Managed Streaming for Apache Kafka)?
A	Amazon MSK is a fully managed Apache Kafka service. Kafka is an open-source distributed event streaming platform. MSK manages cluster provisioning, patching, and replication. Used for real-time data pipelines and streaming applications requiring high throughput and guaranteed message ordering.
□	*Real-World Example: A fintech company uses MSK: 1,000 applications publish transactions to Kafka topics. Fraud detection, analytics, and notification consumers independently read the same topics at their own pace. MSK handles millions of messages/second — MSK removes the operational burden of managing Kafka clusters.*

Q243	What is AWS Lake Formation?
A	AWS Lake Formation builds, secures, and manages data lakes. Automates data ingestion from S3, RDS, DynamoDB. Creates a Glue Data Catalog. Enforces fine-grained access control including column-level and row-level security. Built on S3 and AWS Glue.
▯	*Real-World Example: A company builds a data lake: ingest from 5 RDS databases + S3 ▯ auto-catalogued by Lake Formation ▯ data scientists query with Athena ▯ marketing uses QuickSight. Lake Formation enforces marketing sees only customer demographics (not financial columns) — column-level security with no code changes.*

Section 23 · IoT & Edge

Q244	What is AWS IoT Core?
A	AWS IoT Core is a managed cloud platform for connecting IoT devices to AWS. Devices communicate via MQTT or HTTP. IoT Core routes messages via a rules engine to Lambda, DynamoDB, S3, Kinesis. Supports billions of devices and trillions of messages with no infrastructure to manage.
▯	*Real-World Example: A smart factory has 10,000 sensors publishing readings every second via MQTT to IoT Core. Rules route: temperature>90C ▯ Lambda triggers maintenance alert, all readings ▯ Kinesis ▯ S3 for 3-year historical analysis. Zero servers provisioned for message routing.*

Q245	What is AWS Wavelength?
A	AWS Wavelength embeds AWS compute within 5G networks, enabling ultra-low latency (<10ms) for mobile applications. Use cases: autonomous vehicles, AR/VR, real-time gaming, connected factories.
▯	*Real-World Example: An AR stadium app overlays live player statistics on fans' phone cameras. Without Wavelength: 80ms latency (visible lag). With Wavelength (in the 5G network): <5ms — overlay appears instantaneous. The difference between a compelling product and an unusable one.*

Q246	What is AWS Outposts rack vs servers?
A	AWS Outposts Rack: 42U rack-scale form factor with full AWS compute, storage, database, and networking for large enterprise data centers. AWS Outposts Servers: 1U/2U server form factor for smaller locations (retail stores, branch offices, factories) where a full rack is impractical.

▯	*Real-World Example: A retail chain installs Outposts Servers in each of their 500 stores. Each server runs local inventory management and POS processing with <5ms local latency. All 500 Outposts are managed centrally from AWS Console — same APIs, same tools as the cloud.*

Section 24 · Additional Services

Q247	What is AWS CloudShell?
A	AWS CloudShell is a browser-based shell with AWS CLI pre-installed, accessible from the AWS Console. Runs with your current console session's permissions. Provides 1GB persistent storage per region. Free to use. Eliminates the need to configure AWS CLI locally.
▯	*Real-World Example: An administrator on-call receives an alert on their phone. They open the AWS Console in mobile Safari, launch CloudShell, and run: aws ec2 stop-instances --instance-ids i-0123456789abcdef0. The problematic instance is stopped within 60 seconds — from a phone, without any local tools installed.*

Q248	What is Amazon CodeGuru?
A	Amazon CodeGuru uses ML to find code defects and performance issues. CodeGuru Reviewer integrates with GitHub/CodeCommit to review pull requests for bugs, security issues, and bad practices. CodeGuru Profiler analyses runtime performance, identifying CPU-intensive code paths.
▯	*Real-World Example: A developer submits a PR with an AWS SDK call inside a loop (making 1,000 API calls instead of one batch call). CodeGuru Reviewer automatically comments: 'This pattern causes high latency and costs — use BatchGetItem instead.' Bug caught before code reaches production.*

Q249	What is AWS License Manager?
A	AWS License Manager helps track and manage software licenses from Microsoft, Oracle, SAP, and other vendors across EC2 instances and on-premises. Enforces licensing rules, generates compliance reports, and prevents licence violations.
▯	*Real-World Example: A company has 200 SQL Server licences. License Manager tracks 187 in use. A developer tries to launch SQL Server instance #201 — License Manager blocks the launch: 'Licence limit of 200 reached.' This prevents a compliance violation that could result in a $500K audit finding.*

Q250	What is the AWS Acceptable Use Policy (AUP)?
A	The AWS AUP prohibits use of AWS services for: illegal activities, distributing malware, running DDoS attacks, port scanning, sending spam, child exploitation, and intellectual property violations. All AWS customers must agree to the AUP upon account creation. Violations can result in account termination.

Q251	What is the difference between a public subnet and a private subnet?
A	Public subnet: has a route to an Internet Gateway (0.0.0.0/0 □ IGW) — resources with public IPs can receive internet traffic. Private subnet: no route to IGW — resources are not directly accessible from the internet. Private subnets use NAT Gateway for outbound-only internet access (OS updates, API calls).
□	*Real-World Example: Web servers (need to receive traffic from users) □ public subnet with Security Group allowing 80/443 from internet. Database servers (must not be internet-accessible) □ private subnet, no public IP. Even if an attacker finds the DB's IP, there is no route from the internet to reach it.*

Q252	What is Amazon Elastic IP pricing?
A	Elastic IPs are free while associated with a running EC2 instance. AWS charges $0.005/hour (~$3.60/month) for: EIPs associated with stopped instances, EIPs not associated with any instance, and additional EIPs beyond the first one on a running instance. This pricing encourages efficient IP usage.
□	*Real-World Example: A company has 10 Elastic IPs. 8 are on running instances (free). 2 are unassociated (floating IPs). Cost: 2 × $0.005/hr × 720 hrs = $7.20/month. Lesson: release unused EIPs — they cost money and waste a limited IPv4 address resource.*

Q253	What is Amazon VPC Flow Logs analysis?
A	VPC Flow Logs record accepted and rejected traffic. Log format: version, account-id, interface-id, srcaddr, dstaddr, srcport, dstport, protocol, packets, bytes, start, end, action (ACCEPT/REJECT), log-status. Stored in CloudWatch Logs or S3. Queryable with Athena or CloudWatch Logs Insights.
□	*Real-World Example: Security team investigates: is our database being scanned? Athena query on Flow Logs: SELECT srcaddr, COUNT(*) FROM flows WHERE dstport=3306 AND action='REJECT' AND start>1704067200 GROUP BY srcaddr ORDER BY 2 DESC. Results: IP 198.51.100.5 attempted 5,000 connections to port 3306 — a port scanner detected.*

Q254	What is Amazon S3 Object Lock?
A	S3 Object Lock prevents objects from being deleted or overwritten for a fixed period or indefinitely. Two modes: Governance (users with special permissions can override) and Compliance (no one, including root, can delete during retention period). Used for WORM (Write Once Read Many) compliance requirements.
□	*Real-World Example: A financial firm is required by SEC regulations to retain trade records for 7 years and they must be immutable (no modifications). S3 Object Lock Compliance mode with 7-year retention ensures records cannot be deleted or modified — even by the AWS account root user — satisfying the regulatory requirement.*

Q255	What is Amazon Glacier Vault Lock?
A	Glacier Vault Lock allows you to deploy and enforce compliance controls (like WORM) on S3 Glacier vaults via a vault lock policy. Once locked, the policy cannot be changed. Designed for regulatory compliance where data must be immutable for years or decades.

Q256	What is AWS Server Migration Service (SMS)?
A	AWS Server Migration Service automates migration of on-premises VMware, Hyper-V, and Azure VMs to AWS. SMS creates incremental AMIs from your virtual machines and replicates them to AWS, minimising downtime. Largely superseded by Application Migration Service (MGN) for new migrations.

Q257	What is the difference between IAM User, Group, Role, and Policy?
A	IAM User: individual identity with permanent credentials (username/password, access keys). IAM Group: collection of users sharing the same permissions (e.g., Developers group). IAM Role: identity assumed temporarily by AWS services, apps, or users — no permanent credentials. IAM Policy: JSON document defining permissions, attached to users/groups/roles.
□	*Real-World Example: New hire joins: create IAM User □ add to 'Developers' group (has S3/EC2 read policy) □ they inherit permissions. Lambda function needs S3 access: create IAM Role with S3 write policy □ attach role to Lambda (no credentials in code). Auditor needs temp access: IAM Role with read-only policy □ auditor assumes role for 4 hours.*

Section 25 · Final 43 Questions – Mixed Topics

Q258	What is Amazon SageMaker Canvas?
A	SageMaker Canvas is a no-code ML tool for business analysts. Upload data, SageMaker automatically builds and trains ML models, provides predictions. No coding or ML expertise needed. Uses AutoML under the hood.
◻	*Real-World Example: A retail buyer (non-technical) uses SageMaker Canvas to predict which products will sell out next week. They upload 2 years of sales history, click 'Build Model', wait 2 hours, and get a model that predicts inventory needs with 87% accuracy — no data scientist required.*

Q259	What is Amazon Bedrock?
A	Amazon Bedrock is a fully managed service for building generative AI applications using foundation models (FMs) from AWS and third-party providers (Anthropic Claude, Meta Llama, Stability AI). No infrastructure to manage. Pay-per-token API access. Supports RAG (Retrieval Augmented Generation) via Knowledge Bases.
◻	*Real-World Example: A company builds a customer service chatbot using Bedrock with Claude. The chatbot answers product questions using their product database (RAG). Bedrock handles the model hosting, scaling, and security — the team writes the application logic and connects their data sources.*

Q260	What is Amazon Q?
A	Amazon Q is AWS's generative AI assistant for business. Amazon Q Business helps employees answer questions about internal company data (connects to 40+ data sources). Amazon Q Developer helps developers write, debug, and optimise code inside IDEs. Available in the AWS Console for AWS questions.
◻	*Real-World Example: A developer in VS Code is writing a Lambda function. They ask Amazon Q Developer: 'How do I implement DynamoDB pagination in Python?' Q instantly generates working code with pagination. They also ask: 'Review my code for security issues' — Q identifies hardcoded credentials and suggests using Secrets Manager.*

Q261	What is AWS Audit Manager?
A	AWS Audit Manager continuously audits AWS usage to assess risk and compliance with regulations (PCI DSS, HIPAA, GDPR, SOC 2). Automatically collects evidence from AWS services. Generates audit-ready reports, eliminating manual evidence collection.

▯ *Real-World Example: A company prepares for a SOC 2 audit. Instead of spending 3 months manually gathering evidence (CloudTrail logs, Config rules, IAM policies), Audit Manager automatically collects evidence for all SOC 2 controls over the year. The auditor receives a complete evidence folder — audit preparation reduced from 3 months to 2 weeks.*

Q262	What is Amazon Timestream?
A	Amazon Timestream is a fast, scalable, serverless time-series database for IoT and operational applications. Automatically scales to handle trillions of events per day. Built-in time-series analytics functions. Stores recent data in memory (fast queries) and historical data in magnetic store (cheap).

▯ *Real-World Example: A DevOps platform collects metrics from 100,000 servers every 10 seconds: CPU, memory, disk, network. That's 10 million data points/second. Timestream stores and queries this data efficiently — a query for 'average CPU per server last 24 hours' runs in 2 seconds across 864 million data points.*

Q263	What is Amazon HealthLake?
A	Amazon HealthLake is a HIPAA-eligible service for storing, transforming, and analysing health data at scale. It normalises health data to FHIR (Fast Healthcare Interoperability Resources) format. Uses ML to extract medical entities (diagnoses, medications, procedures) from unstructured clinical notes.

Q264	What is AWS Ground Station?
A	AWS Ground Station is a fully managed ground station as a service for controlling satellite communications and downloading satellite data. No satellite infrastructure to build or manage. Pay per minute of antenna use. Available at 12 global ground station locations.

▯ *Real-World Example: A weather forecasting company operates a remote sensing satellite. AWS Ground Station contacts the satellite as it passes over ground station locations globally, downloads imagery to S3, and feeds it to EC2 instances for processing — the company has no physical ground stations to maintain.*

Q265	What is the AWS Free Tier Always Free tier?
A	AWS Always Free services (no 12-month expiry): Lambda (1 million requests/month), DynamoDB (25GB storage + 25 WCU/RCU), SNS (1 million publishes), SQS (1 million requests), CloudWatch (10 custom metrics), CodeBuild (100 build minutes), AWS Glue (1M objects in Data Catalog).

□	*Real-World Example: A developer builds a personal project: Alexa skill using Lambda (free) □ DynamoDB user data (free) □ SNS notifications (free) □ CloudWatch monitoring (free). Monthly AWS bill: $0.00 — entirely on Always Free tier. This enables developers to keep hobby projects running indefinitely for free.*

Q266	What is an AWS Well-Architected Review process?
A	A Well-Architected Review is a structured conversation between solutions architects and workload owners. They answer questions about the workload design for each pillar, identify risks (HRIs and MRIs), and create an improvement plan. AWS Partner Network partners conduct these reviews for customers.

Q267	What is AWS Cost Categories?
A	AWS Cost Categories allow you to map costs to business dimensions (teams, projects, cost centres) using rules. Rules can be based on accounts, services, tags, charge types, or linked accounts. Appear in Cost Explorer and CUR for cost allocation and chargeback to internal teams.
□	*Real-World Example: A company with 20 AWS accounts defines Cost Categories: map accounts 1-5 to Category='Engineering', accounts 6-10 to 'Marketing', accounts 11-15 to 'Operations'. Cost Explorer now shows total spend by business unit — enabling internal chargeback and departmental budget tracking.*

Q268	What is the difference between AWS Managed Policy and Customer Managed Policy?
A	AWS Managed Policies are created and maintained by AWS (e.g., AmazonS3FullAccess, AmazonEC2ReadOnlyAccess). Automatically updated when AWS adds new services. Customer Managed Policies are created by you — custom permissions tailored to your requirements. Inline Policies are embedded directly in a user/group/role (not reusable).
□	*Real-World Example: Use AWS Managed Policy (AmazonEC2FullAccess) for a developer who needs full EC2 access — AWS maintains it. Create Customer Managed Policy for a Lambda function that needs custom permissions (s3:GetObject on a specific bucket + dynamodb:PutItem on a specific table) — precise least-privilege policy.*

Q269	What is AWS Elastic Disaster Recovery?

A	AWS Elastic Disaster Recovery (EDR) enables fast, reliable recovery of on-premises and cloud-based applications. It continuously replicates servers into a low-cost staging area. During a disaster, you launch recovery instances in minutes. RTO: minutes. RPO: seconds to sub-seconds.
▯	*Real-World Example: A manufacturing company's on-premises ERP system fails completely (ransomware attack). AWS Elastic Disaster Recovery had been continuously replicating the ERP server to AWS staging. They initiate recovery: within 20 minutes, the ERP system is running on AWS with data from 30 seconds before the attack.*

Q270	What is Amazon Chime SDK?
A	The Amazon Chime SDK allows developers to embed real-time audio, video, and messaging directly into their applications. Build custom video conferencing, telehealth, virtual classrooms, and collaboration tools. Pay per minute of audio/video.

Q271	What is the AWS Cloud Value Framework?
A	AWS Cloud Value Framework identifies 4 business value dimensions: 1. Cost Savings (lower TCO vs on-premises). 2. Staff Productivity (less time managing infrastructure, more time innovating). 3. Operational Resilience (higher availability, faster recovery). 4. Business Agility (faster time to market, more experiments).
▯	*Real-World Example: A company migrating to AWS measures: Cost Savings ($2M/year lower TCO), Staff Productivity (+40 hours/month freed from server maintenance ▯ reallocated to product development), Operational Resilience (availability improved from 99.5% to 99.99%), Business Agility (new feature deployment time: 3 months ▯ 1 week).*

Q272	What is Amazon WorkDocs?
A	Amazon WorkDocs is a fully managed secure content collaboration and storage service. Think of it as an enterprise Dropbox/Google Drive on AWS. Features: file storage, version control, sharing, commenting, and Active Directory integration for authentication.

Q273	What is the AWS Marketplace Seller Private Offer?

A	AWS Marketplace Private Offer allows sellers to create custom pricing and terms for specific buyers. A seller can offer a specific customer a custom discount, custom EULA, or custom payment schedule through the Marketplace billing mechanism — ideal for enterprise negotiations.

Q274	What is Amazon Elastic File System vs FSx for Windows?
A	EFS: Linux-only NFS file system, automatically scales, multi-AZ, serverless. FSx for Windows: Windows-native SMB file system with NTFS, Active Directory integration, Windows ACLs — required for Windows workloads. Use EFS for Linux workloads; FSx for Windows for Windows Server applications.
▯	*Real-World Example: A company has 20 Windows EC2 instances running a .NET application that needs shared file storage with Windows ACLs and Active Directory authentication. EFS would not work (Windows/NTFS incompatible). FSx for Windows File Server is the correct choice — it speaks SMB and integrates with their AD domain.*

Q275	What is AWS Service Quotas?
A	AWS Service Quotas is a centralised dashboard for viewing and managing quotas (limits) for AWS services in your account. You can request quota increases, monitor usage vs limits, and set CloudWatch alarms when approaching limits.
▯	*Real-World Example: A DevOps engineer checks Service Quotas before a product launch: EC2 On-Demand vCPUs: 32 (used) / 32 (limit) — at 100%! They request an increase to 200 vCPUs. AWS approves in 24 hours. Without proactive quota monitoring, the product launch would have failed with 'InsufficientCapacity' errors.*

Q276	What is AWS re:Invent?
A	AWS re:Invent is AWS's annual global cloud computing conference held in Las Vegas every November/December. The world's largest AWS event — 60,000+ attendees, 2,500+ sessions, hundreds of new service and feature announcements. Available live streamed and as recordings for free.

Q277	What is Amazon EventBridge Pipes?
A	EventBridge Pipes provides point-to-point integration between event producers and consumers with optional filtering, enrichment (via Lambda/API calls), and transformation. Supported sources: SQS, DynamoDB Streams, Kinesis, MSK, MQ. Reduces integration code.

□	*Real-World Example: DynamoDB Streams fires an event for every database change. Without Pipes: write Lambda to read stream, filter, enrich, forward to SQS. With EventBridge Pipes: configure visually — source: DynamoDB Stream □ filter: only INSERT events □ enrich: call Lambda to add customer data □ target: SQS. Zero code for the integration.*

Q278	What is AWS Resilience Hub?
A	AWS Resilience Hub helps you define, validate, and track the resilience of AWS applications. You define your RTO/RPO targets, Resilience Hub assesses your application against those targets, identifies resiliency gaps (missing Multi-AZ, missing backups), and recommends improvements.

Q279	What is Amazon Pinpoint?
A	Amazon Pinpoint is a communications service for sending targeted, personalised messages to customers across channels: email, SMS, push notifications, voice, and in-app messaging. Supports audience segmentation and campaign analytics.
□	*Real-World Example: A mobile app uses Pinpoint: segment users who haven't opened the app in 30 days □ send a personalised push notification 'We miss you! Here's 20% off.' Track open rates, click rates, and conversions. Campaign costs: $0.001/message for push notifications.*

Q280	What is AWS Transfer Family?
A	AWS Transfer Family provides fully managed SFTP, FTPS, FTP, and AS2 file transfer services backed by S3 or EFS. Business partners can continue using familiar protocols (SFTP) while files are automatically stored in S3. No server management required.
□	*Real-World Example: A healthcare company's insurance partners send patient data files via SFTP every night. Instead of maintaining an SFTP server, they use Transfer Family SFTP endpoint backed by S3. Partners connect to the same SFTP address — they don't know or care that it's AWS S3 underneath.*

Q281	What is Amazon AppFlow?
A	Amazon AppFlow is a fully managed integration service for securely transferring data between AWS services and SaaS applications (Salesforce, Marketo, Zendesk, Slack, SAP). No coding required — configure flows in the console. Supports bi-directional data transfer.

▯	*Real-World Example: A marketing team needs Salesforce leads in Amazon Redshift for analysis. AppFlow connects Salesforce ▯ Redshift with a scheduled flow (every hour). IT sets it up in 30 minutes — no ETL code, no API integration code. Marketing gets fresh lead data in Redshift continuously.*

Q282	What is the AWS Activate program?
A	AWS Activate provides startups with free resources to build and grow on AWS: AWS credits (up to $100,000), training credits, AWS Business Support, and access to technical resources. Two tiers: Activate Founders (early-stage, free) and Activate Portfolio (VC-backed startups).
▯	*Real-World Example: A bootstrapped startup applies for AWS Activate Founders. They receive $1,000 in AWS credits — enough to run their MVP for 6 months at no cost. When they raise a seed round, they upgrade to Activate Portfolio and receive $25,000 in credits, free support, and access to AWS solutions architects.*

Q283	What is Amazon Rekognition Custom Labels?
A	Rekognition Custom Labels allows you to build custom image classification and object detection models using your own images — without ML expertise. Provide labeled training images, Rekognition trains a custom model, you call the API for predictions.
▯	*Real-World Example: A quality control team at a factory needs to detect defective products on an assembly line. They label 500 images of good and defective products. Rekognition Custom Labels trains a model in 2 hours. The model inspects 10 products/second on the production line — replacing a manual visual inspection team.*

Q284	What is Amazon Comprehend Medical?
A	Amazon Comprehend Medical is an NLP service that extracts medical information from unstructured text: patient conditions, medications, dosages, procedures, anatomy, and medical tests. HIPAA-eligible. Processes clinical notes, discharge summaries, and medical records.
▯	*Real-World Example: A hospital has 1 million unstructured clinical notes. Comprehend Medical reads: 'Patient presented with acute MI, started on aspirin 325mg and metoprolol 50mg bid.' It extracts: Condition=Acute MI, Medication=Aspirin, Dose=325mg, Medication=Metoprolol, Dose=50mg, Frequency=twice daily. Used for research and population health analytics.*

Q285	What is AWS DeepComposer?

A	AWS DeepComposer is an educational keyboard device for learning generative AI through music. Developers play a melody; DeepComposer uses ML models (GANs, Transformers) to compose an accompaniment. Includes a cloud-based console with music generation and training capabilities.

Q286	What is Amazon Lookout for Metrics?
A	Amazon Lookout for Metrics uses ML to automatically detect anomalies in business and operational metrics (revenue, user signups, page views, server errors). No ML expertise needed — connect data sources, it learns normal patterns and alerts on deviations with severity and root cause analysis.
▯	*Real-World Example: An e-commerce company sets up Lookout for Metrics on revenue data. On a Tuesday afternoon, revenue drops 65% — immediately flagged as a Critical anomaly. Root cause analysis shows it's correlated with checkout page errors (also anomalous). Engineers fix the checkout bug in 15 minutes — revenue restored.*

Q287	What is Amazon Monitron?
A	Amazon Monitron is an end-to-end system for predictive maintenance of industrial equipment. Includes hardware sensors (Monitron Sensors) and a gateway. Sensors measure vibration and temperature; ML models detect abnormal patterns indicating potential equipment failure.
▯	*Real-World Example: A paper mill installs Monitron sensors on 50 industrial motors. ML detects that motor #12 shows unusual vibration patterns — predicted failure in 2 weeks. Maintenance team is notified. They schedule a planned replacement during the next maintenance window, preventing an unplanned outage that would cost $50,000/hour.*

Q288	What is Amazon Kendra vs OpenSearch for enterprise search?
A	OpenSearch is keyword-based search — returns documents containing specific words. Kendra is ML-powered semantic search — understands the intent of natural language questions and returns precise answers from documents. Kendra reads documents; OpenSearch indexes them. For enterprise FAQ-style search, Kendra provides dramatically better user experience.
▯	*Real-World Example: Employee asks: 'How many vacation days do I get after 5 years?' OpenSearch returns: 20 documents mentioning 'vacation'. Kendra reads the HR policy and returns: 'After 5 years of employment, you are entitled to 20 days of annual leave, as stated in the HR Benefits Guide page 12.' Direct answer vs list of documents.*

Q289	What is the CLF-C02 exam passing score?
A	The AWS Certified Cloud Practitioner CLF-C02 exam requires a score of 700 out of 1000 to pass. The exam uses scaled scoring (not simple percentage). Scores range from 100-1000. There is no penalty for wrong answers — always guess rather than leave blank. Results are typically immediate for online exams.
▯	*Real-World Example: Exam strategy: 700/1000 passing = approximately 70% correct answers on scored questions. With 50 scored questions, you need approximately 35 correct. The 15 unscored pilot questions do not count for or against you — you cannot identify which ones they are, so treat all 65 questions equally.*

Q290	What domains are heaviest on the CLF-C02 exam?
A	Domain 3 (Cloud Technology and Services) is the heaviest at 34% — expect ~22 questions on EC2, S3, RDS, Lambda, VPC, CloudFront, Route 53, and other core services. Domain 2 (Security and Compliance) is second at 30% — ~20 questions on IAM, encryption, Shield, GuardDuty, Shared Responsibility.
▯	*Real-World Example: Study priority: Domain 3 (34%) ▯ Core services breadth. Domain 2 (30%) ▯ Security depth. Domain 1 (24%) ▯ Cloud concepts and benefits. Domain 4 (12%) ▯ Pricing and support plans. Don't neglect Domain 4 — 8 questions on pricing/support can be the difference between passing and failing.*

Q291	What is the difference between SSD and HDD EBS volumes?
A	SSD-backed EBS: gp3 (general purpose — boot volumes, web servers, dev environments), io2 (provisioned IOPS — high-performance databases, SAP HANA). HDD-backed EBS: st1 (throughput optimised — big data, log processing, data warehouses — sequential access), sc1 (cold HDD — infrequently accessed, lowest cost).
▯	*Real-World Example: MySQL database: gp3 (balanced cost and performance, 3,000 IOPS guaranteed). High-frequency trading database: io2 (100,000 IOPS, consistent sub-millisecond latency). Nightly ETL that reads 2TB sequentially: st1 (cheap, high sequential throughput). 7-year compliance archive on EBS: sc1 (lowest cost per GB).*

Q292	What is Amazon S3 Replication?

A	S3 replication automatically copies objects across buckets. Cross-Region Replication (CRR): replicates to a bucket in a different region — for compliance, disaster recovery, and latency reduction. Same-Region Replication (SRR): replicates within the same region — for log aggregation, test/prod sync. Requires versioning.

Q293	What is AWS DataSync vs Storage Gateway?
A	AWS DataSync: automated, high-speed data transfer agent for one-time or scheduled migrations/synchronisation between on-premises and AWS (S3, EFS, FSx). Storage Gateway: ongoing hybrid storage integration — extends on-premises storage to AWS, caches frequently accessed data locally. DataSync = data migration; Storage Gateway = hybrid storage integration.
▯	*Real-World Example: DataSync: 200TB historical archive files ▯ migrate to S3 one-time (fast, network-optimised transfer). Storage Gateway File Gateway: 50 on-premises servers that daily write reports ▯ report files transparently stored in S3 (ongoing hybrid integration). Different tools for different jobs.*

Q294	What is AWS Backup Vault Lock?
A	AWS Backup Vault Lock adds WORM (Write Once Read Many) protection to backup vaults. Once enabled, backup jobs cannot be deleted or modified for the configured retention period — even by the account root user. Used for ransomware protection and regulatory compliance.
▯	*Real-World Example: A company is hit by ransomware. Attackers gain admin access and attempt to delete all AWS Backup recovery points to prevent restoration. AWS Backup Vault Lock blocks all deletion attempts — backup recovery points are immutable. IT restores all systems from yesterday's backup. No ransom paid.*

Q295	What is Amazon CloudFront signed URLs and signed cookies?
A	CloudFront signed URLs restrict access to individual files (one URL = one file access). Signed cookies restrict access to multiple files without changing URLs. Both require signature generated with a private key. Used for premium content (paid video streaming, software downloads) that should not be publicly accessible.
▯	*Real-World Example: A premium video streaming service delivers content via CloudFront. Each user's browser gets a signed cookie (valid 8 hours, one per session) that authorises access to all videos in their subscription tier. Without the signed cookie, CloudFront returns 403 Forbidden — preventing content sharing or scraping.*

Q296	What is Amazon Route 53 health checks?
A	Route 53 health checks monitor the health of endpoints (EC2, ELB, on-premises) by sending HTTP/HTTPS/TCP requests every 10-30 seconds. Used with Failover routing to automatically divert traffic away from unhealthy endpoints. Can monitor other health checks (calculated health checks).
▯	*Real-World Example: An application runs in us-east-1 with a failover to eu-west-1. Route 53 health check pings the us-east-1 ALB every 30 seconds. The ALB becomes unhealthy (3 consecutive failures in 90 seconds). Route 53 automatically updates DNS to point to eu-west-1. Users experience 90 seconds of potential issues, then automatic recovery.*

Q297	What is Amazon Inspector vs GuardDuty?
A	Amazon Inspector scans EC2 instances and container images for software vulnerabilities (CVEs) and misconfigured networks — it is a vulnerability scanning tool (what is wrong with my resources). GuardDuty detects active threats and suspicious behaviour from logs — it is a threat detection tool (is something bad happening right now). Both are security services with different focuses.
▯	*Real-World Example: Inspector: EC2 instance running OpenSSL 1.0.2 with CVE-2016-6304 (critical vulnerability) — patch it. GuardDuty: EC2 instance is making DNS requests to a known malware command-and-control domain — it may already be compromised. Inspector = proactive vulnerability management; GuardDuty = reactive threat detection.*

Q298	What is Amazon Macie vs Inspector?
A	Macie: focuses on sensitive DATA in S3 — discovers and classifies PII (credit cards, SSNs, passports) and alerts on anomalous access patterns. Inspector: focuses on INFRASTRUCTURE vulnerabilities on EC2 and containers — software CVEs, network exposure. Macie protects your data; Inspector protects your compute.
▯	*Real-World Example: Macie alert: S3 bucket 'employee-hr-records' contains 10,000 files with SSNs and is publicly accessible — data exposure risk. Inspector alert: EC2 instance has Apache Log4j 2.14.1 (Log4Shell vulnerability CVE-2021-44228) — infrastructure vulnerability. Different tools, different concerns, both essential.*

Q299	What is the principle of defence in depth on AWS?

A	Defence in depth applies multiple layers of security controls so that if one layer fails, others continue to protect the system. On AWS: Edge (WAF + CloudFront + Shield) □ Network (VPC + NACLs + Security Groups) □ Compute (IAM roles + OS hardening) □ Application (input validation) □ Data (encryption at rest + in transit) □ Monitoring (CloudTrail + GuardDuty).
□	*Real-World Example: An attacker bypasses WAF (layer 1 failed). VPC NACLs block their IP range (layer 2 succeeds). Even if NACLs missed them, Security Groups only allow port 443 (layer 3 succeeds). Even if they reached the app, data is encrypted with KMS (layer 5 succeeds). CloudTrail logs all their actions (layer 6). Multiple layers = significantly harder attack.*

Q300	What is Amazon Elastic Kubernetes Service (EKS) Anywhere?
A	EKS Anywhere allows you to create and manage Kubernetes clusters on your own infrastructure (on-premises, bare metal, VMware vSphere) using the same EKS tooling used in AWS. Provides a consistent Kubernetes experience across cloud and on-premises. Uses EKS Distro (the same Kubernetes distribution used in AWS).
□	*Real-World Example: A company has strict data sovereignty laws requiring some workloads on-premises. They run EKS Anywhere on their VMware vSphere cluster using the same kubectl commands, Helm charts, and CI/CD pipelines as their cloud EKS clusters. Operations teams learn one set of tools for both environments — dramatically reducing complexity.*

Thank You for Using TechShiksha!

This guide covers all domains of the AWS Certified Cloud Practitioner (CLF-C02) exam.
Practice these 300 questions to build confidence for your certification journey.

▯ Website	akhileshchaudhari.com
▯ Instagram	@techshiksha_official
▯ LinkedIn	TechShiksha
▯ Telegram	t.me/techshiksha